

Executive Summary — Surrogate Model Validation Report

Use case: UCLOADS\_1 | 07 September 2026 | 5 model(s)

REQUIREMENTS PARTIALLY MET

Recommended model: PyTorchNN

Avg R<sup>2</sup>: 0.9923 — Excellent

Q90 Accuracy — No output passes

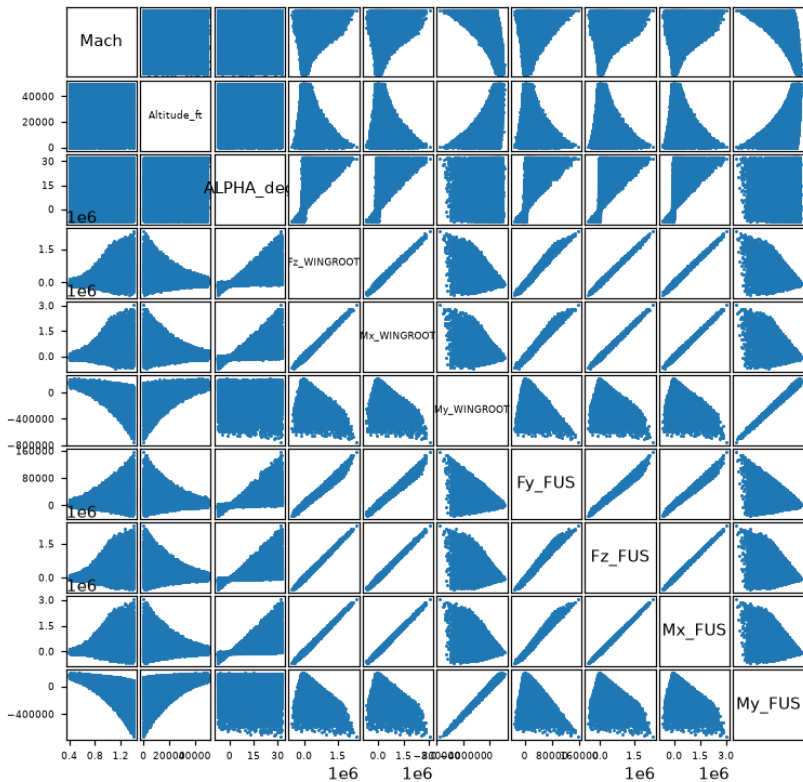
Fail: Fz\_WINGROOT, Mx\_WINGROOT, My\_WINGROOT, Fy\_FUS, Fz\_FUS, Mx\_FUS, My\_FUS, Mz\_FUS

Project Overview

|                    |                                                                                  |
|--------------------|----------------------------------------------------------------------------------|
| Use case           | UCLOADS_1                                                                        |
| Inputs             | Mach, Altitude_ft, ALPHA_deg                                                     |
| Outputs            | 8: Fz_WINGROOT, Mx_WINGROOT, My_WINGROOT, Fy_FUS, Fz_FUS, Mx_FUS, My_FUS, Mz_FUS |
| Train / Val / Test | 70% / 10% / 20%                                                                  |
| Accuracy target    | Q90 < 10% relative error per output                                              |

Variable Correlation — Input vs Output

Variable Correlation



Data Statistics

Per-variable statistics over all data (train + val + test), as `Train_set.describe()` reports them in the feature-selection notebook: count, mean, standard deviation, minimum, quartiles and maximum.

Inputs

|             | count  | mean       | std        | min    | P25        | P50        | P75        | max        |
|-------------|--------|------------|------------|--------|------------|------------|------------|------------|
| Mach        | 30,000 | 0.8993     | 0.2877     | 0.4    | 0.649      | 0.899      | 1.148      | 1.4        |
| Altitude_ft | 30,000 | 25041.4981 | 14450.3288 | 0.262  | 12538.2808 | 24908.6162 | 37629.5684 | 49999.7461 |
| ALPHA_deg   | 30,000 | 12.6492    | 11.542     | -7.499 | 2.7138     | 12.705     | 22.6283    | 32.5       |

Input variables — Train\_set.describe()

Outputs

|             | count  | mean        | std         | min          | P25         | P50         | P75         | max         |
|-------------|--------|-------------|-------------|--------------|-------------|-------------|-------------|-------------|
| Fz_WINGROOT | 30,000 | 170703.0851 | 330033.1836 | -552389.4375 | -29380.1304 | 59228.2832  | 255270.5    | 2,310,000   |
| Mx_WINGROOT | 30,000 | 219817.2595 | 433912.0613 | -719582.25   | -41977.8672 | 72184.8438  | 328885.5859 | 3,050,000   |
| My_WINGROOT | 30,000 | 23493.4845  | 148166.7046 | -754026.4375 | -29575.6533 | 74652.375   | 127117.2227 | 216981.3906 |
| Fy_FUS      | 30,000 | 13347.9878  | 18600.438   | -30868.4082  | 2188.8232   | 7538.3909   | 18198.4678  | 156598.9844 |
| Fz_FUS      | 30,000 | 170836.2941 | 329666.0959 | -526906.0625 | -28808.978  | 55768.1953  | 253443.3555 | 2,300,000   |
| Mx_FUS      | 30,000 | 219365.6596 | 433865.0107 | -683626.6875 | -43647.6865 | 66522.2109  | 324146.1953 | 3,030,000   |
| My_FUS      | 30,000 | 23581.2207  | 148229.718  | -726381.75   | -28888.1357 | 75282.5625  | 127646.9004 | 189144.7969 |
| Mz_FUS      | 30,000 | -22660.0748 | 27871.8468  | -263049.875  | -25834.228  | -12596.9438 | -7368.7069  | 7201.8691   |

Output variables — Train\_set.describe()

Model Selection

| Model            | Avg R <sup>2</sup> | Q90 pass (8) | KS pass (8) |
|------------------|--------------------|--------------|-------------|
| MLP              | 0.9917             | 0/8          | 8/8         |
| GradientBoosting | 0.9907             | 0/8          | 7/8         |
| RandomForest     | 0.9917             | 0/8          | 0/8         |
| XGBoost          | 0.9590             | 0/8          | 7/8         |
| PyTorchNN ★      | 0.9923             | 0/8          | 8/8         |

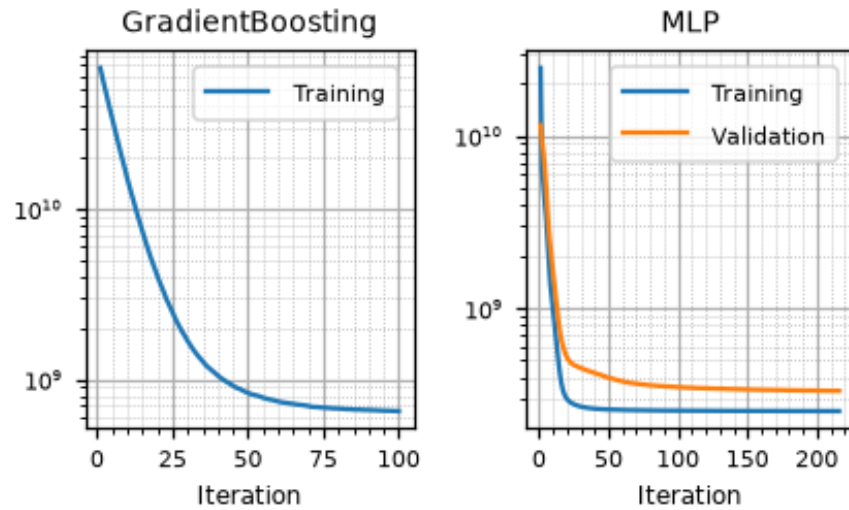
★ Recommended model. Q90 target: < 10%. KS:  $p \geq 0.05$  = no overfitting.

Accuracy Assessment (Q90 & R<sup>2</sup>)

| Output      | MLP            |        |         | GradientBoosting |        |         | RandomForest   |        |         | XGBoost        |        |         | PyTorchNN ★    |        |         |
|-------------|----------------|--------|---------|------------------|--------|---------|----------------|--------|---------|----------------|--------|---------|----------------|--------|---------|
|             | R <sup>2</sup> | Q90    | Gap     | R <sup>2</sup>   | Q90    | Gap     | R <sup>2</sup> | Q90    | Gap     | R <sup>2</sup> | Q90    | Gap     | R <sup>2</sup> | Q90    | Gap     |
| Fz_WINGROOT | 0.9900         | 0.2474 | +0.1474 | 0.9880           | 0.2706 | +0.1706 | 0.9894         | 0.2555 | +0.1555 | 0.9452         | 0.5830 | +0.4830 | 0.9901         | 0.2475 | +0.1475 |
| Mx_WINGROOT | 0.9900         | 0.2501 | +0.1501 | 0.9881           | 0.2733 | +0.1733 | 0.9892         | 0.2596 | +0.1596 | 0.9452         | 0.5863 | +0.4863 | 0.9901         | 0.2494 | +0.1494 |
| My_WINGROOT | 0.9900         | 0.2052 | +0.1052 | 0.9896           | 0.2096 | +0.1096 | 0.9892         | 0.2131 | +0.1131 | 0.9898         | 0.2075 | +0.1075 | 0.9901         | 0.2042 | +0.1042 |
| Fy_FUS      | 0.9876         | 0.2364 | +0.1364 | 0.9879           | 0.2344 | +0.1344 | 0.9896         | 0.2180 | +0.1180 | 0.9462         | 0.4800 | +0.3800 | 0.9901         | 0.2133 | +0.1133 |
| Fz_FUS      | 0.9975         | 0.1262 | +0.0262 | 0.9954           | 0.1676 | +0.0676 | 0.9972         | 0.1314 | +0.0314 | 0.9523         | 0.5560 | +0.4560 | 0.9975         | 0.1250 | +0.0250 |
| Mx_FUS      | 0.9975         | 0.1274 | +0.0274 | 0.9954           | 0.1695 | +0.0695 | 0.9973         | 0.1319 | +0.0319 | 0.9523         | 0.5596 | +0.4596 | 0.9975         | 0.1258 | +0.0258 |
| My_FUS      | 0.9974         | 0.1038 | +0.0038 | 0.9970           | 0.1122 | +0.0122 | 0.9969         | 0.1121 | +0.0121 | 0.9972         | 0.1075 | +0.0075 | 0.9975         | 0.1023 | +0.0023 |
| Mz_FUS      | 0.9836         | 0.2595 | +0.1595 | 0.9840           | 0.2566 | +0.1566 | 0.9851         | 0.2485 | +0.1485 | 0.9437         | 0.4527 | +0.3527 | 0.9859         | 0.2423 | +0.1423 |

R<sup>2</sup>≥0.95  R<sup>2</sup>∈ [0.80, 0.95)  R<sup>2</sup><0.80 or Q90 fail  pass Gap=Q90−target (<0 ⇒ margin)

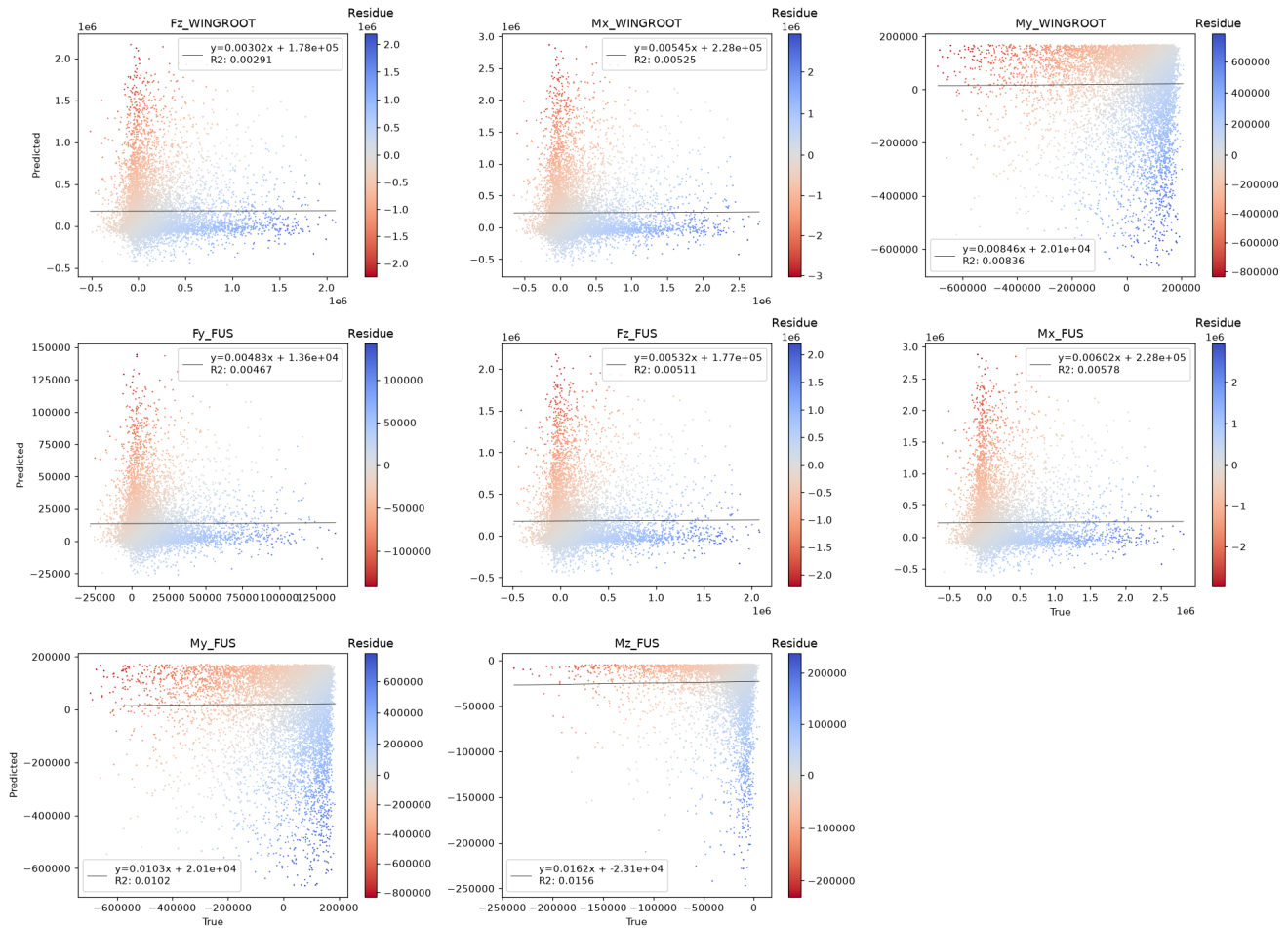
## Training History



Training and validation loss per iteration, log scale. For models trained with early stopping, the validation loss is approximated from the recorded validation  $R^2$  via  $(1 - R^2) \cdot \text{Var}(y)/2$ . Gradient boosting reports per-stage deviance averaged over its per-output estimators.

## Predicted vs True — PyTorchNN

Scatter plot [True vs Predicted]



## Data Split Quality

| Metric                                        | Value | Pass |
|-----------------------------------------------|-------|------|
| Residual voxel proportion ( $\leq 0.05$ )     | 0.000 | ✓    |
| Valid test proportion                         | 0.966 |      |
| p-hacking test proportion ( $\leq 0.05$ warn) | 0.005 | ✓    |
| Isolated test proportion ( $\leq 0.05$ warn)  | 0.029 | ✓    |
| Chi <sup>2</sup> p-value ( $\geq 0.05$ )      | —     | —    |

No p-hacking concern: only 0.5% of test points lie closer to a training point than training points are to each other, so the test error is not flattered.

## Overfitting Check (KS Test) — PyTorchNN

| Output      | KS p-value |
|-------------|------------|
| Fz_WINGROOT | 0.891      |
| Mx_WINGROOT | 0.172      |
| My_WINGROOT | 0.663      |
| Fy_FUS      | 0.586      |
| Fz_FUS      | 0.501      |
| Mx_FUS      | 0.523      |
| My_FUS      | 0.791      |
| Mz_FUS      | 0.479      |

■  $p \geq 0.05$  — no overfitting ■  $p < 0.05$  — overfitting detected

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# Deep Analysis

UCLOADS\_1

Winner Model — Full Validation

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## Deep Analysis — PyTorchNN

Mirrors `validation_template.ipynb` — all computations from validation CSVs

- 3.1 Data Overview
- 3.2 Train-Test Split Analysis
- 3.3 Error Quantification —  $P(E)$
- 3.4  $P(E|X)$  — Error Conditional on Inputs
- 3.5  $P(E|Y)$  — Error Conditional on Outputs
- 3.6 Uncertainty Analysis

## Data Overview

Statistical summary of inputs and outputs over all data (train + val + test), as reported by `describe()` in the feature-selection notebook: count, mean, standard deviation, minimum, quartiles and maximum per variable. All figures in this part are drawn by `validationlib`, matching the extended validation notebook.

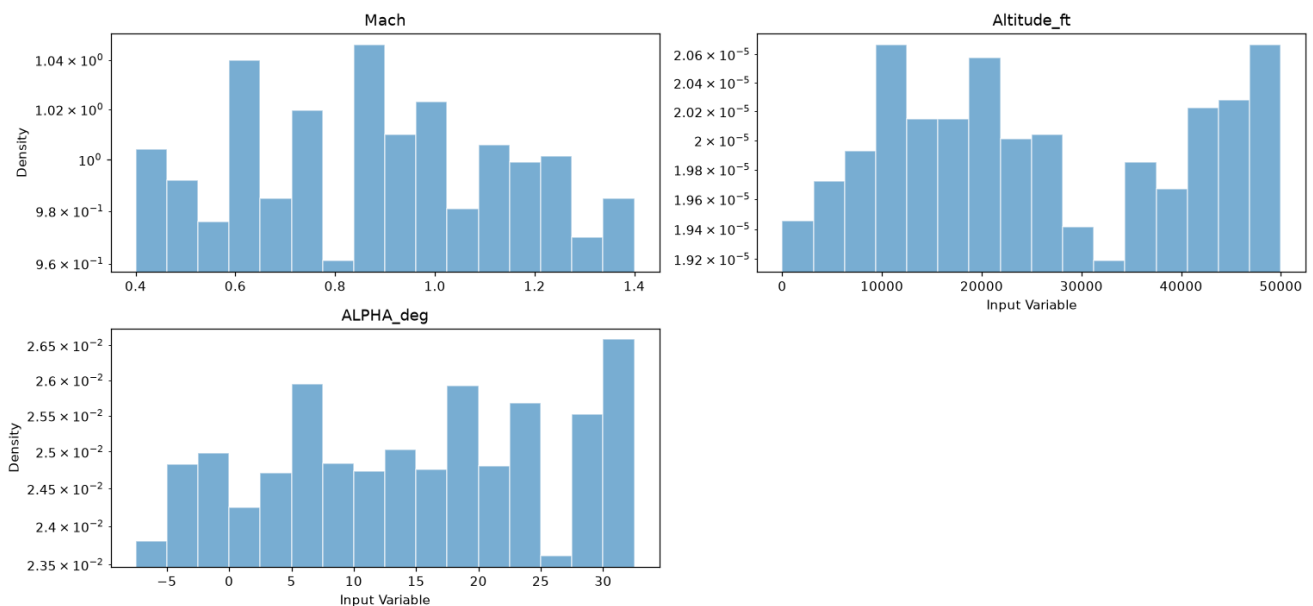
## Input Statistics

|             | count  | mean       | std        | min    | P25        | P50        | P75        | max        |
|-------------|--------|------------|------------|--------|------------|------------|------------|------------|
| Mach        | 30,000 | 0.8993     | 0.2877     | 0.4    | 0.649      | 0.899      | 1.148      | 1.4        |
| Altitude_ft | 30,000 | 25041.4981 | 14450.3288 | 0.262  | 12538.2808 | 24908.6162 | 37629.5684 | 49999.7461 |
| ALPHA_deg   | 30,000 | 12.6492    | 11.542     | -7.499 | 2.7138     | 12.705     | 22.6283    | 32.5       |

Input variables — `Train_set.describe()`

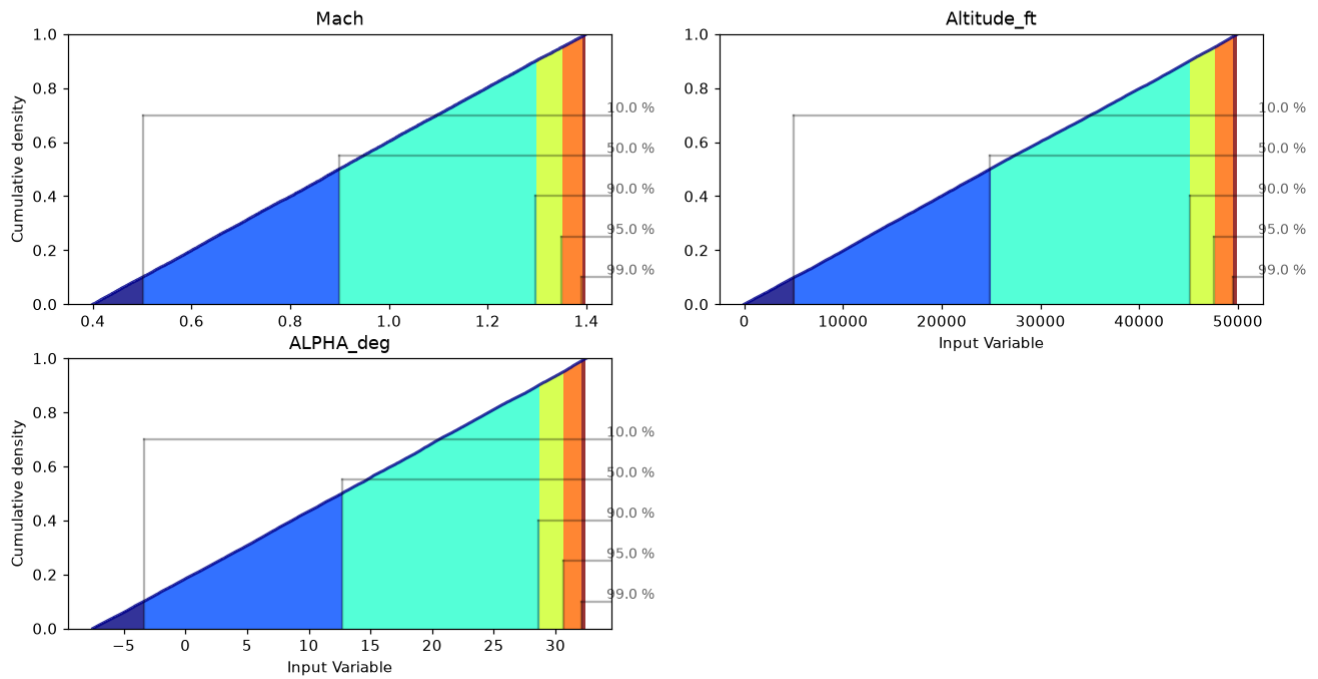
## Input Histograms

### Histogram of [Input Variable] inside $\mu \pm 3\sigma$



## Input Cumulative Distributions

## Cumulative plot of [Input Variable]



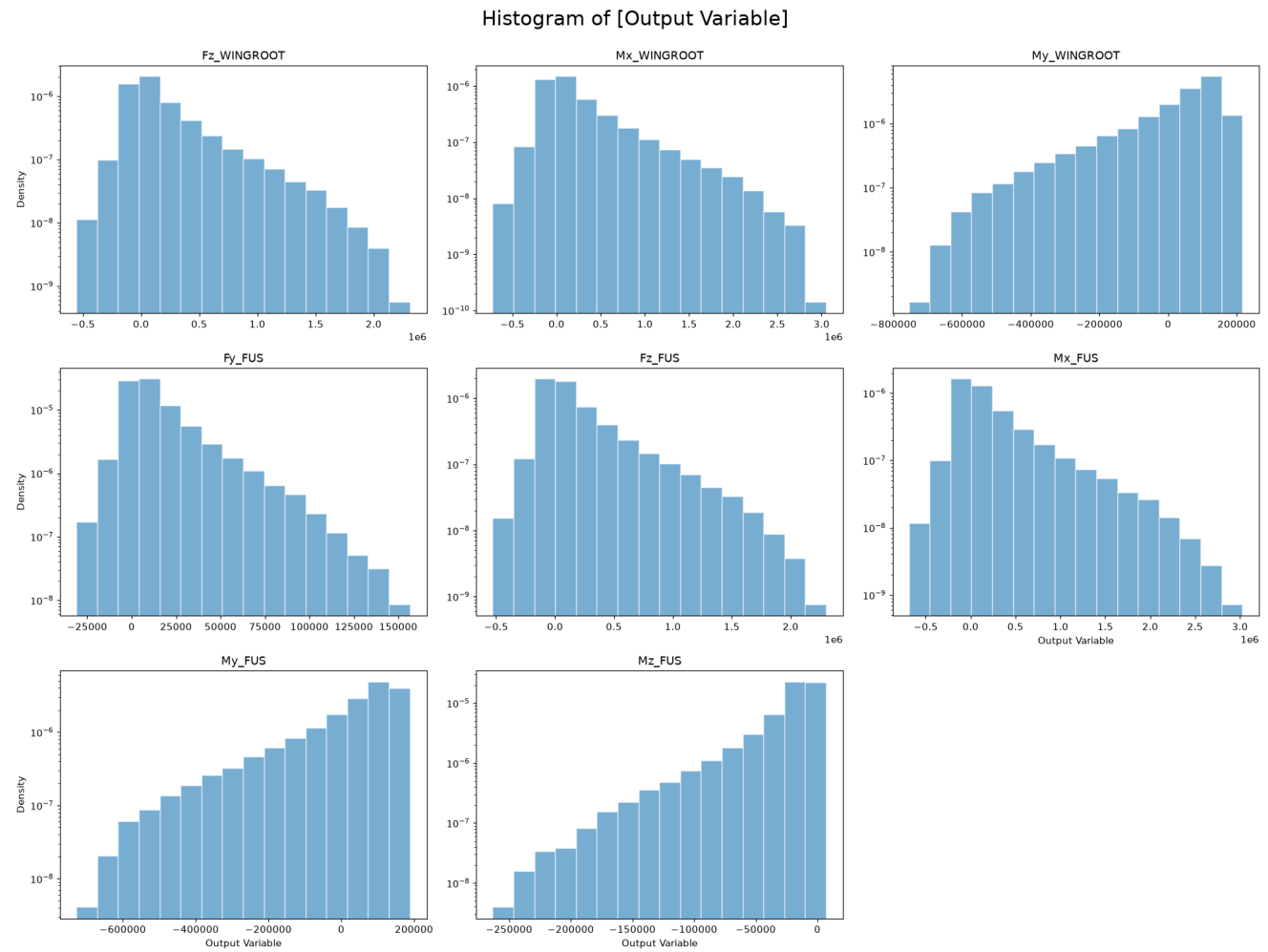
## Output Statistics

|             | count  | mean        | std         | min          | P25         | P50         | P75         | max         |
|-------------|--------|-------------|-------------|--------------|-------------|-------------|-------------|-------------|
| Fz_WINGROOT | 30,000 | 170703.0851 | 330033.1836 | -552389.4375 | -29380.1304 | 59228.2832  | 255270.5    | 2,310,000   |
| Mx_WINGROOT | 30,000 | 219817.2595 | 433912.0613 | -719582.25   | -41977.8672 | 72184.8438  | 328885.5859 | 3,050,000   |
| My_WINGROOT | 30,000 | 23493.4845  | 148166.7046 | -754026.4375 | -29575.6533 | 74652.375   | 127117.2227 | 216981.3906 |
| Fy_FUS      | 30,000 | 13347.9878  | 18600.438   | -30868.4082  | 2188.8232   | 7538.3909   | 18198.4678  | 156598.9844 |
| Fz_FUS      | 30,000 | 170836.2941 | 329666.0959 | -526906.0625 | -28808.978  | 55768.1953  | 253443.3555 | 2,300,000   |
| Mx_FUS      | 30,000 | 219365.6596 | 433865.0107 | -683626.6875 | -43647.6865 | 66522.2109  | 324146.1953 | 3,030,000   |
| My_FUS      | 30,000 | 23581.2207  | 148229.718  | -726381.75   | -28888.1357 | 75282.5625  | 127646.9004 | 189144.7969 |
| Mz_FUS      | 30,000 | -22660.0748 | 27871.8468  | -263049.875  | -25834.228  | -12596.9438 | -7368.7069  | 7201.8691   |

Output variables — Train\_set.describe()

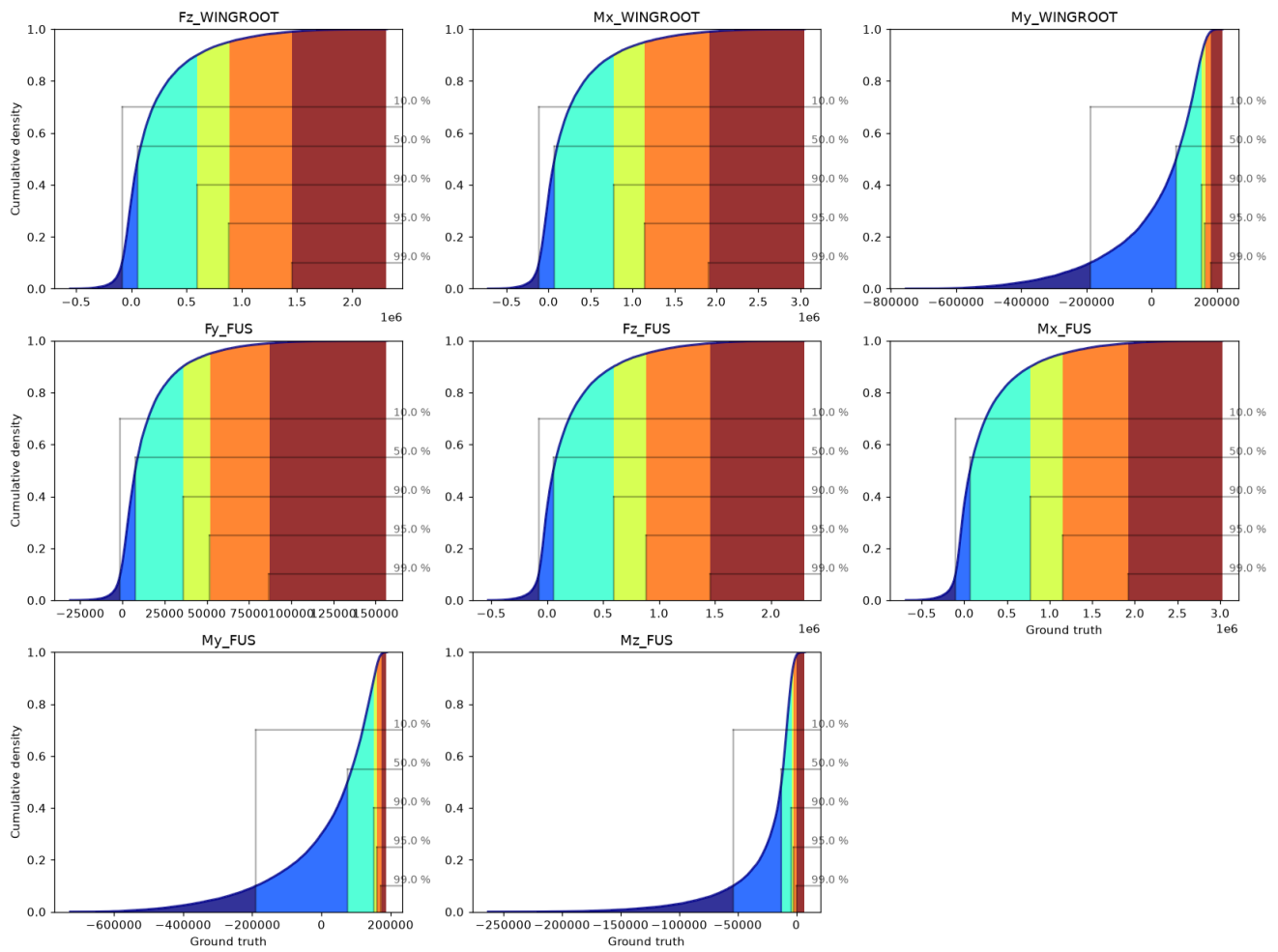


## Output Histograms



## Output Cumulative Distributions

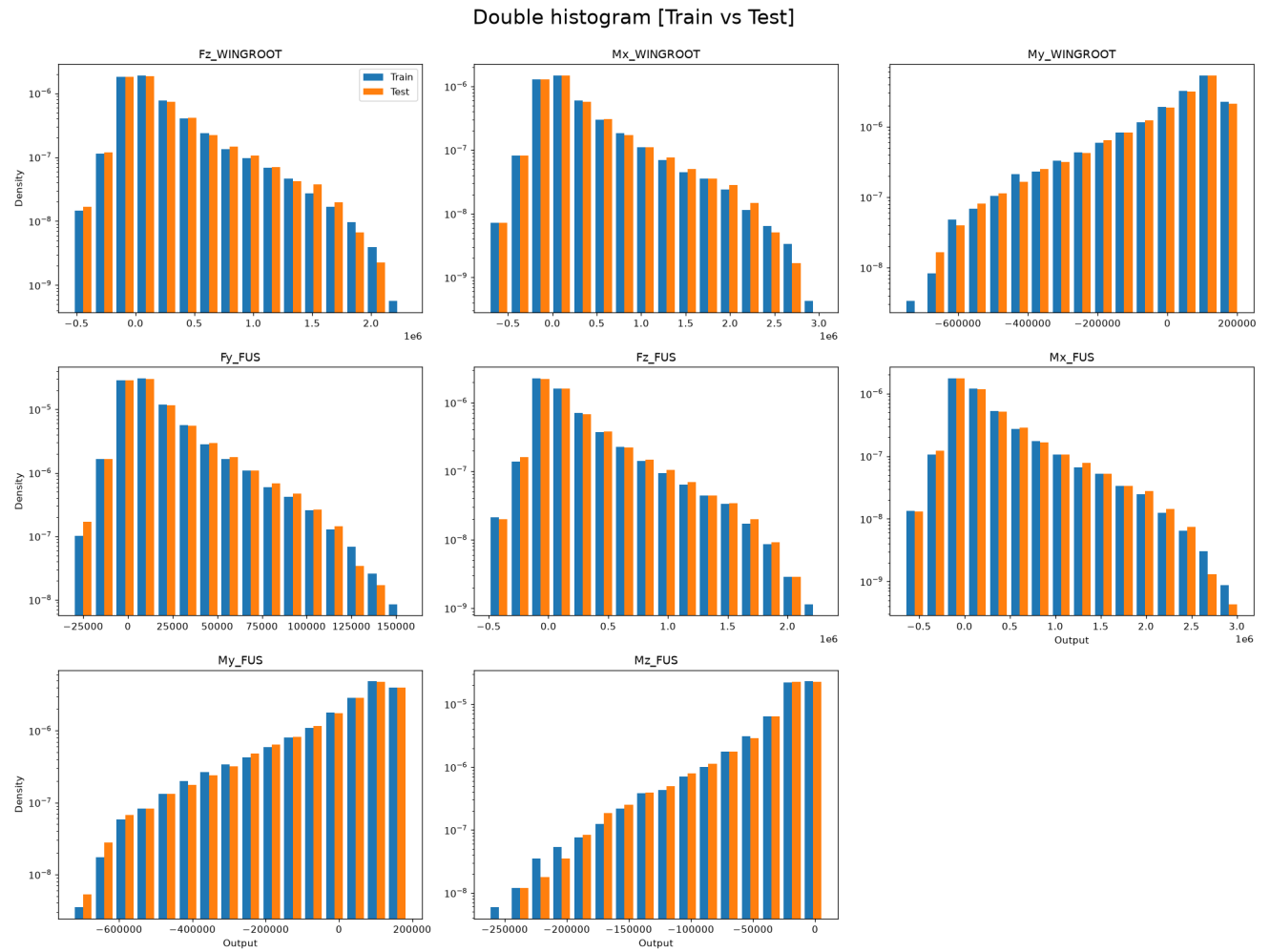
## Cumulative plot of [Ground truth]



## Train-Test Split Analysis

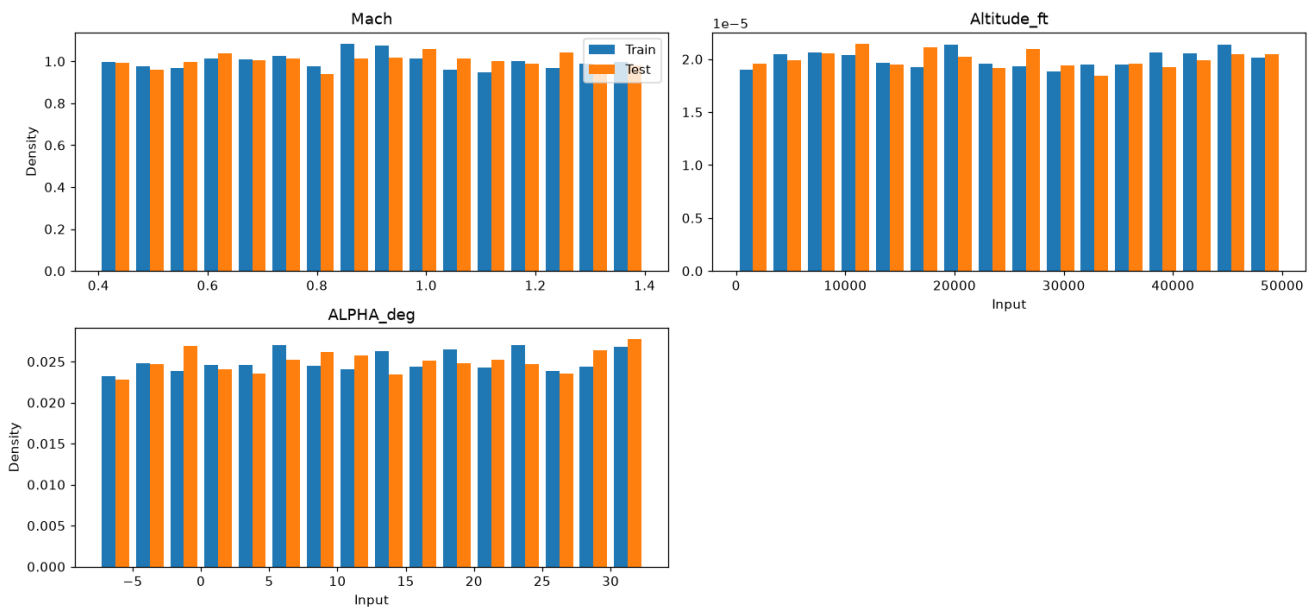
A well-designed split produces training and test distributions that are similar but not identical. The Kolmogorov-Smirnov (KS) test and the Anderson-Darling (AD) test are used to assess distributional similarity.  $H_0$ : both sets come from the same distribution.  $p < 0.05$  rejects  $H_0$  (undesirable if too low; undesirable if the split was done by stratification and  $p$  is too high).

### Output Distributions: Train vs Test



## Input Distributions: Train vs Test

Double histogram [Train vs Test]



## KS and AD Test Results

|             | KS     | AD   |
|-------------|--------|------|
| Fz_WINGROOT | 0.6637 | 0.25 |
| Mx_WINGROOT | 0.8629 | 0.25 |
| My_WINGROOT | 0.7802 | 0.25 |
| Fy_FUS      | 0.8335 | 0.25 |
| Fz_FUS      | 0.8534 | 0.25 |
| Mx_FUS      | 0.9062 | 0.25 |
| My_FUS      | 0.4894 | 0.25 |
| Mz_FUS      | 0.7912 | 0.25 |

Train vs test p-values. Red:  $p < 0.05$  (distributions differ).

## Split Quality — Voxel Tessellation Proximity (p-hacking check)

Beyond matching distributions, a split can still flatter the model if test points sit right next to training points (p-hacking) or probe regions the model never saw (isolated, residual voxel). The VTPM method from `validationlib` (SF\_9 cell 9.0) classifies every test point.

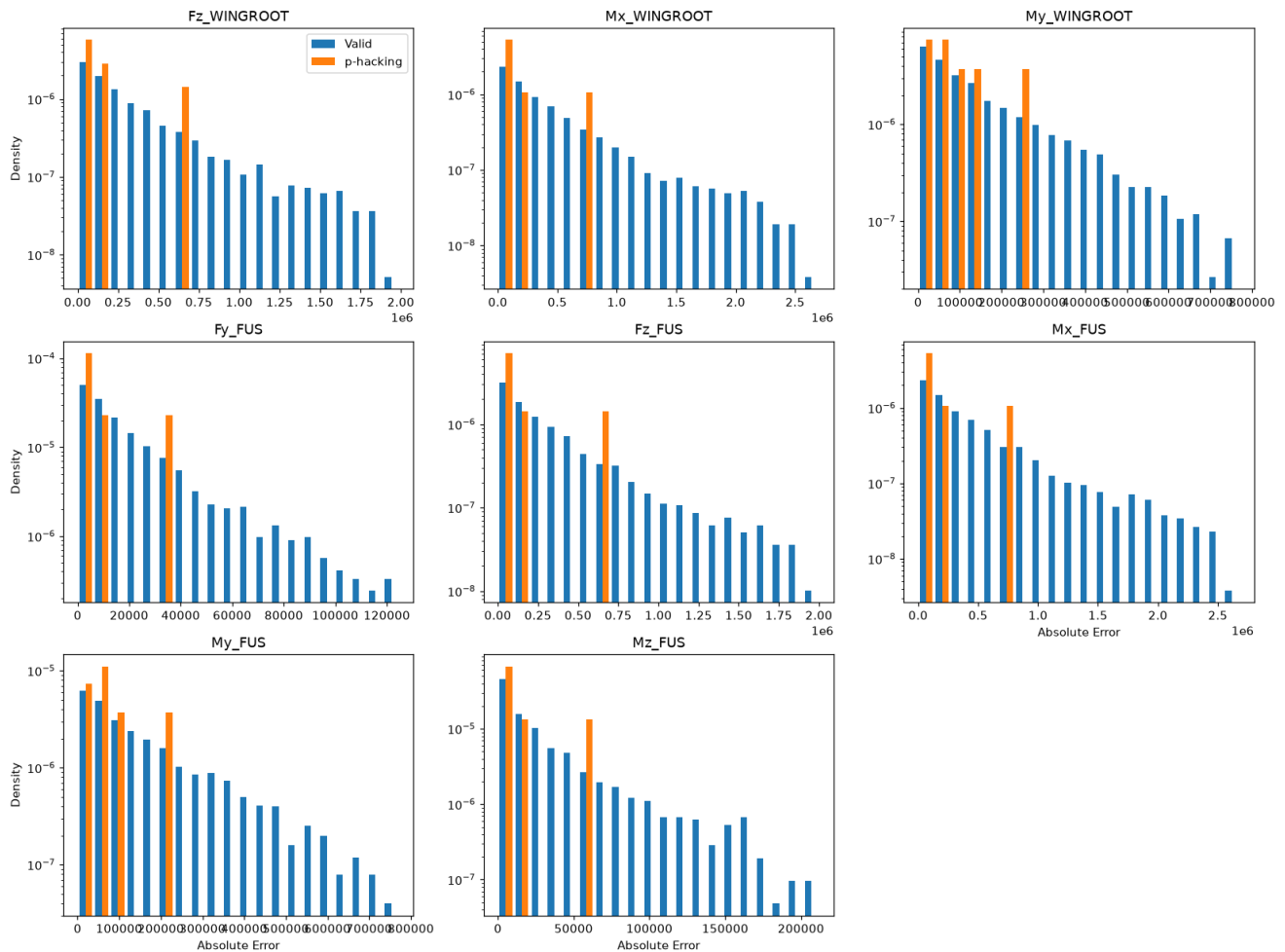
| Metric                                        | Value | Pass |
|-----------------------------------------------|-------|------|
| Residual voxel proportion ( $\leq 0.05$ )     | 0.000 | ✓    |
| Valid test proportion                         | 0.966 |      |
| p-hacking test proportion ( $\leq 0.05$ warn) | 0.005 | ✓    |
| Isolated test proportion ( $\leq 0.05$ warn)  | 0.029 | ✓    |
| Chi <sup>2</sup> p-value ( $\geq 0.05$ )      | —     | —    |

No p-hacking concern: only 0.5% of test points lie closer to a training point than training points are to each other, so the test error is not flattered.

## Error Distribution: Valid vs p-hacking Test Points

Absolute error of the test points flagged as p-hacking against the valid ones, exactly as the extended validation notebook draws it. Matching distributions (high p-values) mean the flagged points do not distort the reported error.

## Double histogram [Valid vs p-hacking]



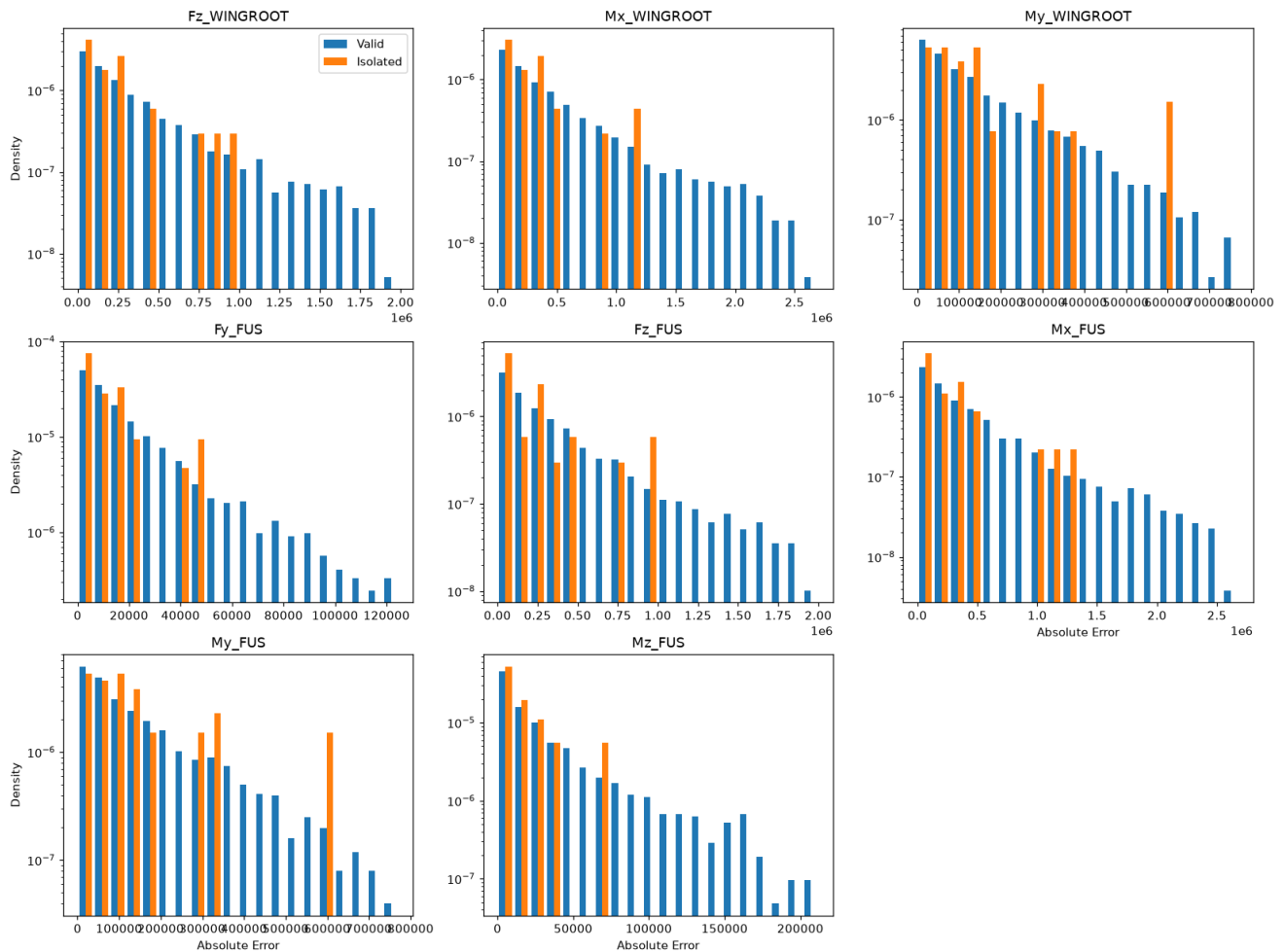
|             | AD     | KS     | KW     |
|-------------|--------|--------|--------|
| Fz_WINGROOT | 0.25   | 0.1981 | 0.1892 |
| Mx_WINGROOT | 0.0324 | 0.0764 | 0.0342 |
| My_WINGROOT | 0.25   | 0.4799 | 0.23   |
| Fy_FUS      | 0.1025 | 0.0688 | 0.0688 |
| Fz_FUS      | 0.0124 | 0.0221 | 0.02   |
| Mx_FUS      | 0.0166 | 0.0688 | 0.0295 |
| My_FUS      | 0.25   | 0.3994 | 0.2023 |
| Mz_FUS      | 0.0324 | 0.0213 | 0.0449 |

Valid ( $n = 1959$ ) vs p-hacking ( $n = 7$ ) absolute-error distributions, on a subsample of the test set. Red:  $p < 0.05$  (the flagged points behave differently).

## Error Distribution: Valid vs Isolated Test Points

Same comparison for the test points with no training neighbour.

## Double histogram [Valid vs Isolated]



|             | AD     | KS     | KW     |
|-------------|--------|--------|--------|
| Fz_WINGROOT | 0.1027 | 0.036  | 0.0657 |
| Mx_WINGROOT | 0.0609 | 0.0494 | 0.0455 |
| My_WINGROOT | 0.25   | 0.6467 | 0.9086 |
| Fy_FUS      | 0.0744 | 0.0607 | 0.0589 |
| Fz_FUS      | 0.0636 | 0.0778 | 0.0459 |
| Mx_FUS      | 0.0836 | 0.0829 | 0.0561 |
| My_FUS      | 0.25   | 0.574  | 0.9619 |
| Mz_FUS      | 0.25   | 0.4272 | 0.3739 |

Valid ( $n = 1959$ ) vs Isolated ( $n = 34$ ) absolute-error distributions, on a subsample of the test set. Red:  $p < 0.05$  (the flagged points behave differently).

## Error Quantification — P(E)

Two error metrics are studied:

- Residue:  $e_i = y_i - \hat{y}_i$ . A centred ( $\mu \approx 0$ ), symmetric distribution indicates an unbiased model.
- Absolute Error:  $|e_i|$ . Always non-negative; its Q90 must satisfy the accuracy requirement.

## Residue Statistics Table

Bootstrap confidence bounds shown as superscript (upper) and subscript (lower).

|             | count  | min        | max       | mean   | (BS)mean                          | median | (BS)median                       | std     | (BS)std                             | IQR     | (BS)IQR                             | kurtosis | (BS)kurtosis                    | skewness | (BS)skewness                          |
|-------------|--------|------------|-----------|--------|-----------------------------------|--------|----------------------------------|---------|-------------------------------------|---------|-------------------------------------|----------|---------------------------------|----------|---------------------------------------|
| Fz_WINGROOT | 10,000 | -2,250,000 | 2,190,000 | -4,860 | <sup>5220</sup> <sub>-13700</sub> | -1,120 | <sup>6000</sup> <sub>-6550</sub> | 478,000 | <sup>489000</sup> <sub>469000</sub> | 395,000 | <sup>410000</sup> <sub>383000</sub> | 2.53     | <sup>2.74</sup> <sub>2.34</sub> | -0.104   | <sup>-0.004</sup> <sub>-0.203</sub>   |
| Mx_WINGROOT | 10,000 | -3,030,000 | 2,920,000 | -7,070 | <sup>4390</sup> <sub>-20200</sub> | -885   | <sup>5010</sup> <sub>-7850</sub> | 628,000 | <sup>640000</sup> <sub>617000</sub> | 516,000 | <sup>530000</sup> <sub>503000</sub> | 2.61     | <sup>2.84</sup> <sub>2.4</sub>  | -0.112   | <sup>-0.0124</sup> <sub>-0.223</sub>  |
| My_WINGROOT | 10,000 | -834,000   | 780,000   | 2,170  | <sup>5800</sup> <sub>-1910</sub>  | 2,370  | <sup>5270</sup> <sub>-473</sub>  | 210,000 | <sup>213000</sup> <sub>206000</sub> | 201,000 | <sup>206000</sup> <sub>195000</sub> | 1.39     | <sup>1.53</sup> <sub>1.26</sub> | -0.0099  | <sup>0.0486</sup> <sub>-0.0901</sub>  |
| Fy_FUS      | 10,000 | -141,000   | 143,000   | -244   | <sup>163</sup> <sub>-745</sub>    | -66.1  | <sup>187</sup> <sub>-376</sub>   | 26,800  | <sup>27600</sup> <sub>26200</sub>   | 22,400  | <sup>23600</sup> <sub>21600</sub>   | 3.17     | <sup>3.47</sup> <sub>2.83</sub> | -0.114   | <sup>0.009</sup> <sub>-0.235</sub>    |
| Fz_FUS      | 10,000 | -2,220,000 | 2,200,000 | -4,950 | <sup>7280</sup> <sub>-15600</sub> | 213    | <sup>6060</sup> <sub>-4780</sub> | 478,000 | <sup>490000</sup> <sub>468000</sub> | 395,000 | <sup>405000</sup> <sub>384000</sub> | 2.55     | <sup>2.74</sup> <sub>2.38</sub> | -0.103   | <sup>-0.0012</sup> <sub>-0.193</sub>  |
| Mx_FUS      | 10,000 | -2,980,000 | 2,940,000 | -6,810 | <sup>6230</sup> <sub>-15900</sub> | 489    | <sup>5600</sup> <sub>-4110</sub> | 629,000 | <sup>641000</sup> <sub>617000</sub> | 513,000 | <sup>526000</sup> <sub>497000</sub> | 2.63     | <sup>2.8</sup> <sub>2.42</sub>  | -0.105   | <sup>-0.0032</sup> <sub>-0.202</sub>  |
| My_FUS      | 10,000 | -828,000   | 780,000   | 2,340  | <sup>6390</sup> <sub>-1660</sub>  | 2,520  | <sup>5310</sup> <sub>-131</sub>  | 210,000 | <sup>214000</sup> <sub>206000</sub> | 200,000 | <sup>207000</sup> <sub>195000</sub> | 1.41     | <sup>1.53</sup> <sub>1.29</sub> | -0.0125  | <sup>0.0519</sup> <sub>-0.0953</sub>  |
| Mz_FUS      | 10,000 | -230,000   | 238,000   | 469    | <sup>1190</sup> <sub>-548</sub>   | 26.8   | <sup>350</sup> <sub>-190</sub>   | 40,400  | <sup>41500</sup> <sub>39500</sub>   | 23,000  | <sup>23800</sup> <sub>22100</sub>   | 5.22     | <sup>5.64</sup> <sub>4.74</sub> | 0.188    | <sup>0.361</sup> <sub>-0.000973</sub> |

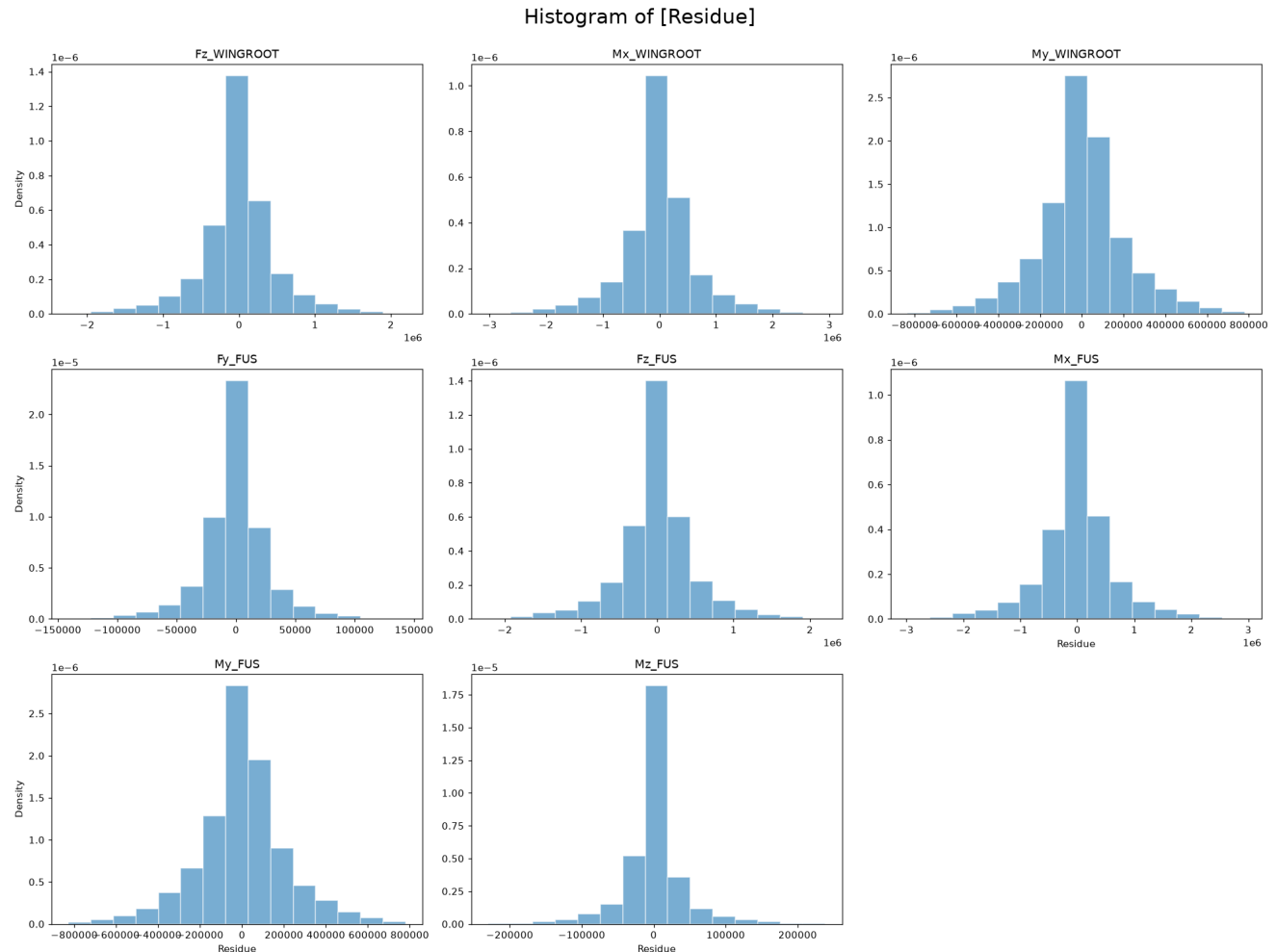
Residue statistics with bootstrap confidence intervals (superscript/subscript bounds).

## Absolute Error Statistics Table

|             | count  | min   | max       | mean    | (BS)mean                            | median  | (BS)median                          | std     | (BS)std                             | IQR     | (BS)IQR                             | kurtosis | (BS)kurtosis                    | skewness | (BS)skewness                    |
|-------------|--------|-------|-----------|---------|-------------------------------------|---------|-------------------------------------|---------|-------------------------------------|---------|-------------------------------------|----------|---------------------------------|----------|---------------------------------|
| Fz_WINGROOT | 10,000 | 42.6  | 2,250,000 | 325,000 | <sup>332000</sup> <sub>319000</sub> | 198,000 | <sup>205000</sup> <sub>192000</sub> | 351,000 | <sup>360000</sup> <sub>344000</sub> | 362,000 | <sup>375000</sup> <sub>351000</sub> | 3.44     | <sup>3.78</sup> <sub>3.07</sub> | 1.82     | <sup>1.87</sup> <sub>1.76</sub> |
| Mx_WINGROOT | 10,000 | 1.58  | 3,030,000 | 424,000 | <sup>433000</sup> <sub>416000</sub> | 258,000 | <sup>265000</sup> <sub>250000</sub> | 463,000 | <sup>472000</sup> <sub>451000</sub> | 472,000 | <sup>485000</sup> <sub>464000</sub> | 3.53     | <sup>3.94</sup> <sub>3.12</sub> | 1.84     | <sup>1.89</sup> <sub>1.77</sub> |
| My_WINGROOT | 10,000 | 26.1  | 834,000   | 150,000 | <sup>153000</sup> <sub>147000</sub> | 100,000 | <sup>103000</sup> <sub>97400</sub>  | 147,000 | <sup>149000</sup> <sub>144000</sub> | 175,000 | <sup>181000</sup> <sub>170000</sub> | 1.98     | <sup>2.2</sup> <sub>1.76</sub>  | 1.48     | <sup>1.52</sup> <sub>1.42</sub> |
| Fy_FUS      | 10,000 | 1.61  | 143,000   | 18,100  | <sup>18500</sup> <sub>17800</sub>   | 11,200  | <sup>11500</sup> <sub>10900</sub>   | 19,800  | <sup>20300</sup> <sub>19200</sub>   | 19,400  | <sup>19900</sup> <sub>18800</sub>   | 4.79     | <sup>5.19</sup> <sub>4.23</sub> | 2.02     | <sup>2.09</sup> <sub>1.94</sub> |
| Fz_FUS      | 10,000 | 12    | 2,220,000 | 324,000 | <sup>332000</sup> <sub>317000</sub> | 197,000 | <sup>205000</sup> <sub>192000</sub> | 352,000 | <sup>361000</sup> <sub>344000</sub> | 364,000 | <sup>378000</sup> <sub>353000</sub> | 3.43     | <sup>3.81</sup> <sub>3.13</sub> | 1.82     | <sup>1.88</sup> <sub>1.77</sub> |
| Mx_FUS      | 10,000 | 26.5  | 2,980,000 | 424,000 | <sup>433000</sup> <sub>414000</sub> | 256,000 | <sup>264000</sup> <sub>249000</sub> | 465,000 | <sup>473000</sup> <sub>452000</sub> | 476,000 | <sup>492000</sup> <sub>468000</sub> | 3.52     | <sup>3.88</sup> <sub>3.24</sub> | 1.84     | <sup>1.91</sup> <sub>1.79</sub> |
| My_FUS      | 10,000 | 12.6  | 828,000   | 150,000 | <sup>153000</sup> <sub>147000</sub> | 99,800  | <sup>104000</sup> <sub>97500</sub>  | 147,000 | <sup>150000</sup> <sub>145000</sub> | 175,000 | <sup>180000</sup> <sub>171000</sub> | 1.98     | <sup>2.2</sup> <sub>1.75</sub>  | 1.48     | <sup>1.52</sup> <sub>1.43</sub> |
| Mz_FUS      | 10,000 | 0.821 | 238,000   | 24,400  | <sup>25000</sup> <sub>23900</sub>   | 11,500  | <sup>12100</sup> <sub>11200</sub>   | 32,200  | <sup>33100</sup> <sub>31400</sub>   | 26,100  | <sup>27300</sup> <sub>25000</sub>   | 6.42     | <sup>7.1</sup> <sub>5.71</sub>  | 2.37     | <sup>2.46</sup> <sub>2.28</sub> |

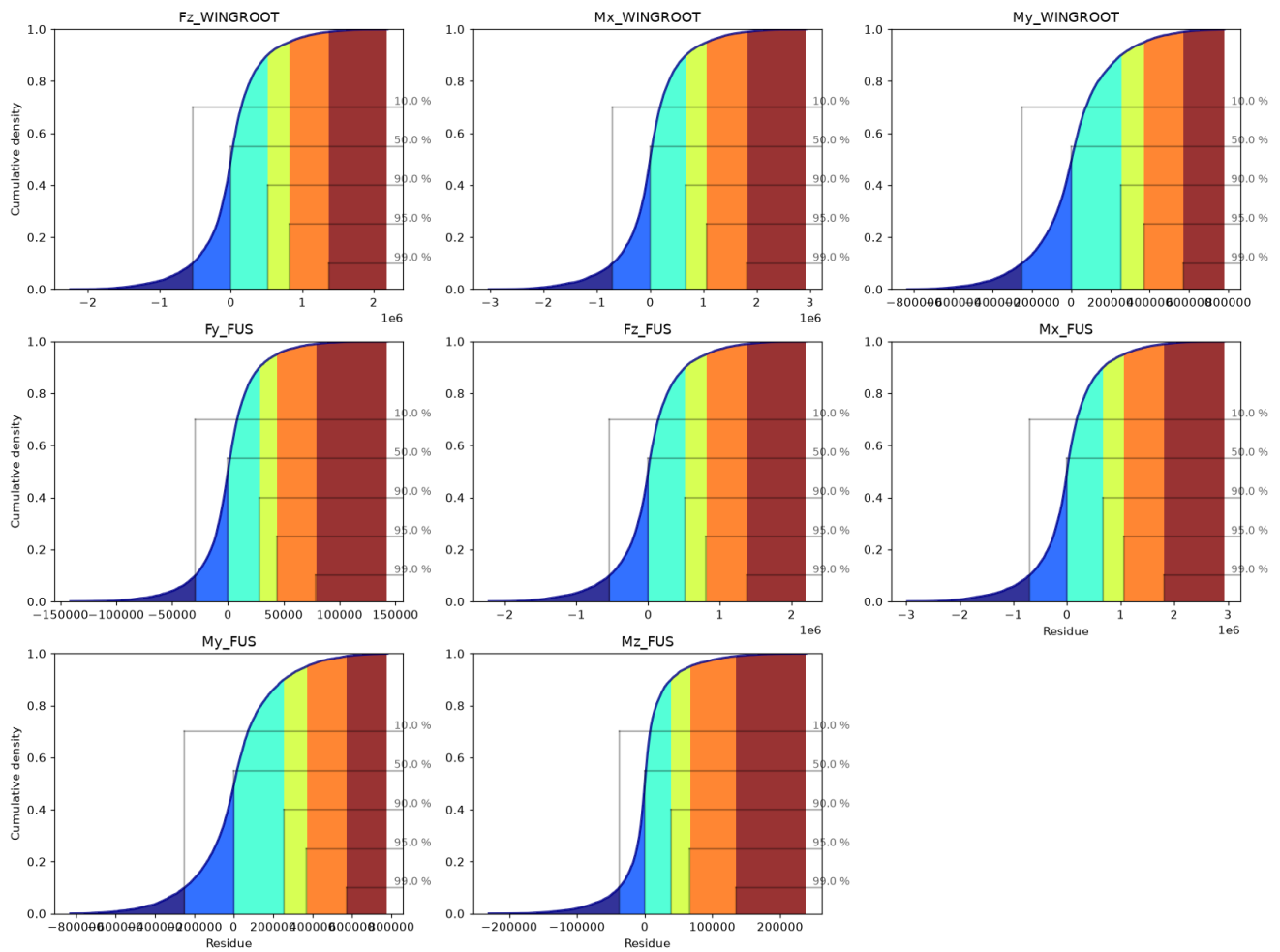
Absolute-error statistics with bootstrap confidence intervals.

## Residue Histograms



## Residue CDF

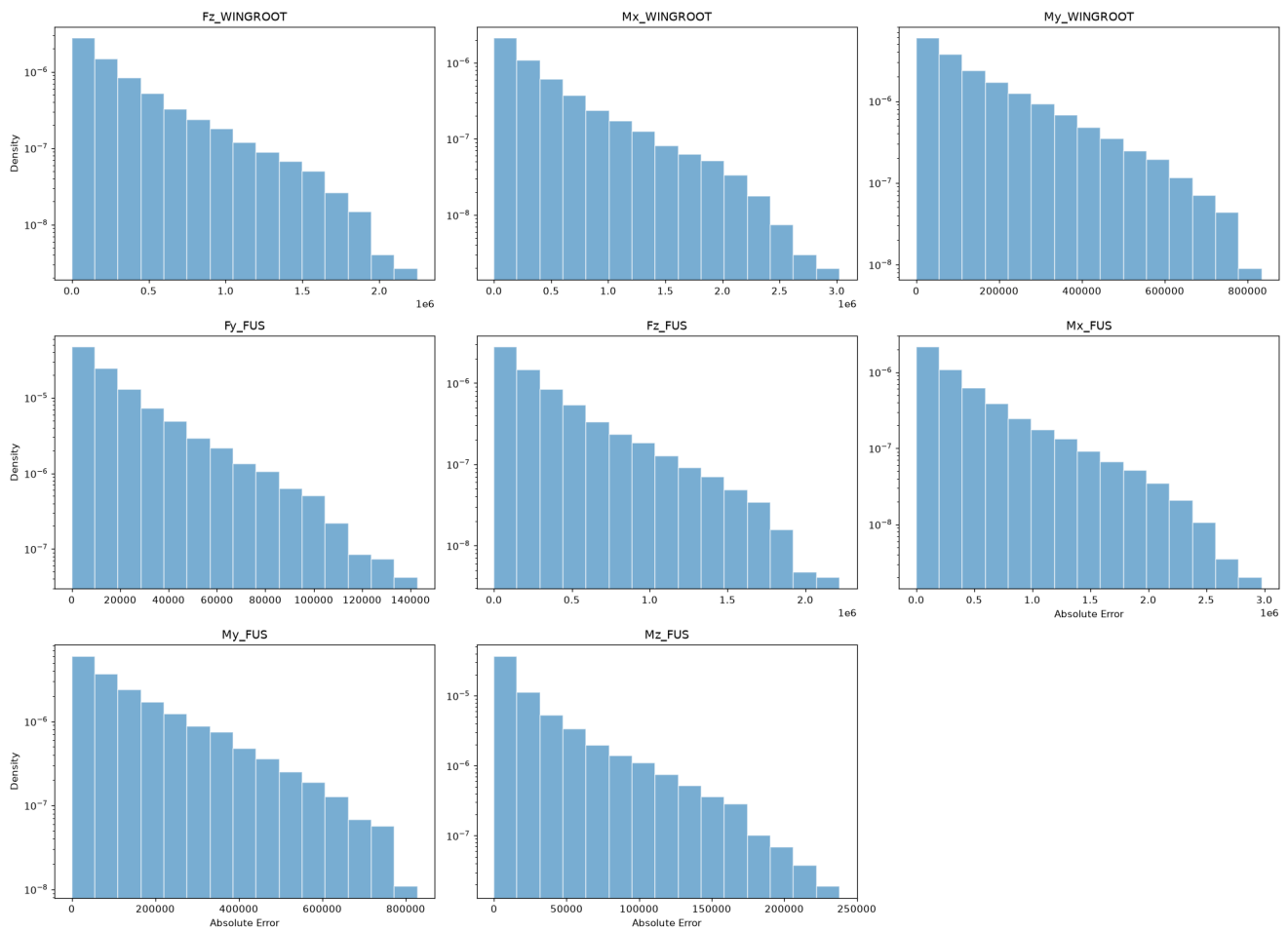
Cumulative plot of [Residue]





## Absolute Error Histograms

Histogram of [Absolute Error]



## Q90 Accuracy Requirement

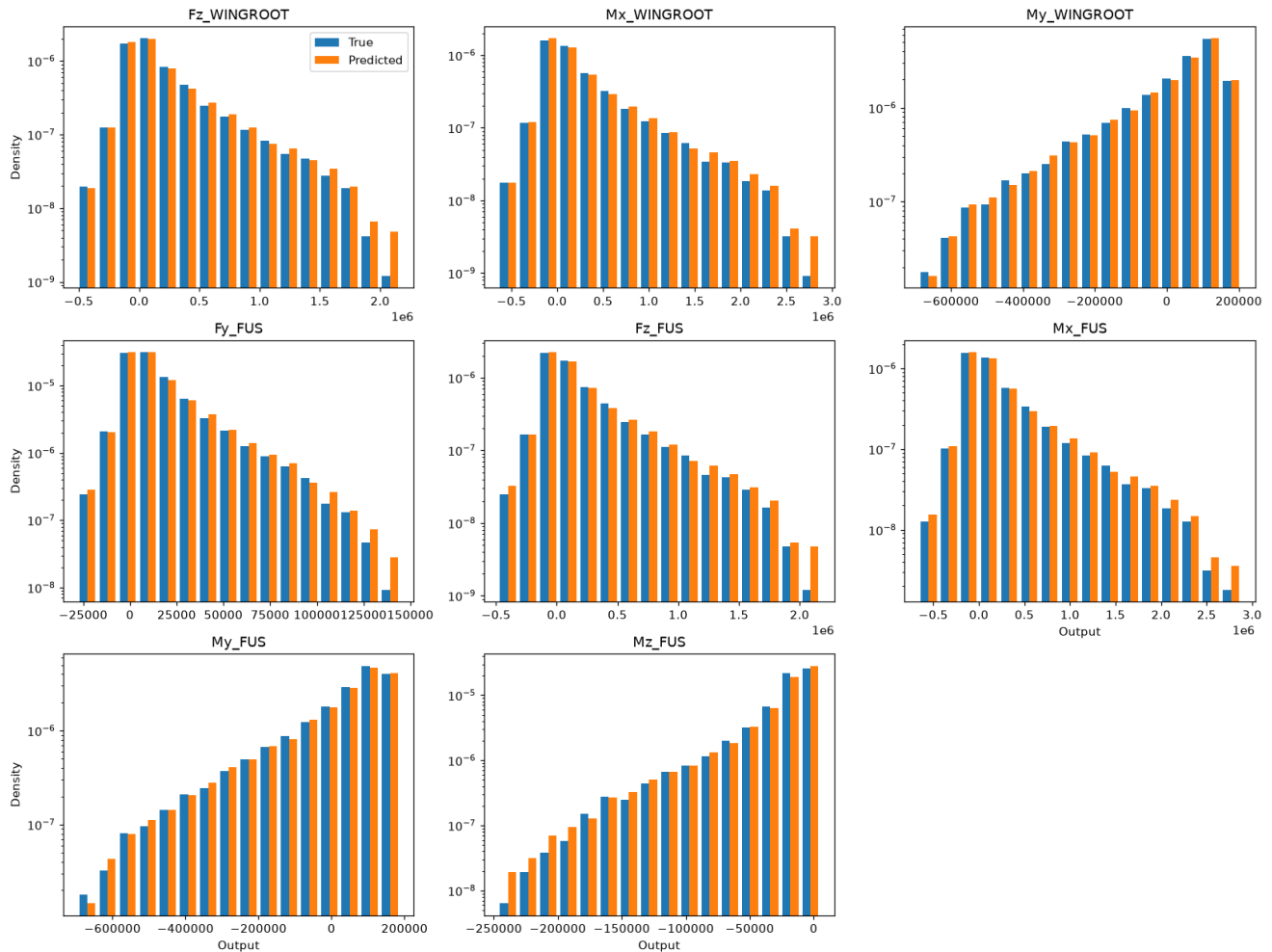
| Output      | Q90     | Target | Status |
|-------------|---------|--------|--------|
| Fz_WINGROOT | 12.8856 | 10%    | FAIL   |
| Mx_WINGROOT | 12.8081 | 10%    | FAIL   |
| My_WINGROOT | 4.9689  | 10%    | FAIL   |
| Fy_FUS      | 10.8875 | 10%    | FAIL   |
| Fz_FUS      | 13.1123 | 10%    | FAIL   |
| Mx_FUS      | 13.3095 | 10%    | FAIL   |
| My_FUS      | 4.8529  | 10%    | FAIL   |
| Mz_FUS      | 5.7248  | 10%    | FAIL   |

■  $p \geq 0.05$  — pass
 ■  $p < 0.05$  — fail

## True vs Predicted Distribution Comparison

AD and KS tests comparing the distribution of true vs. predicted values on the test set. A significant difference indicates the model is not reproducing the output distribution faithfully.

## Double histogram [True vs Predicted]

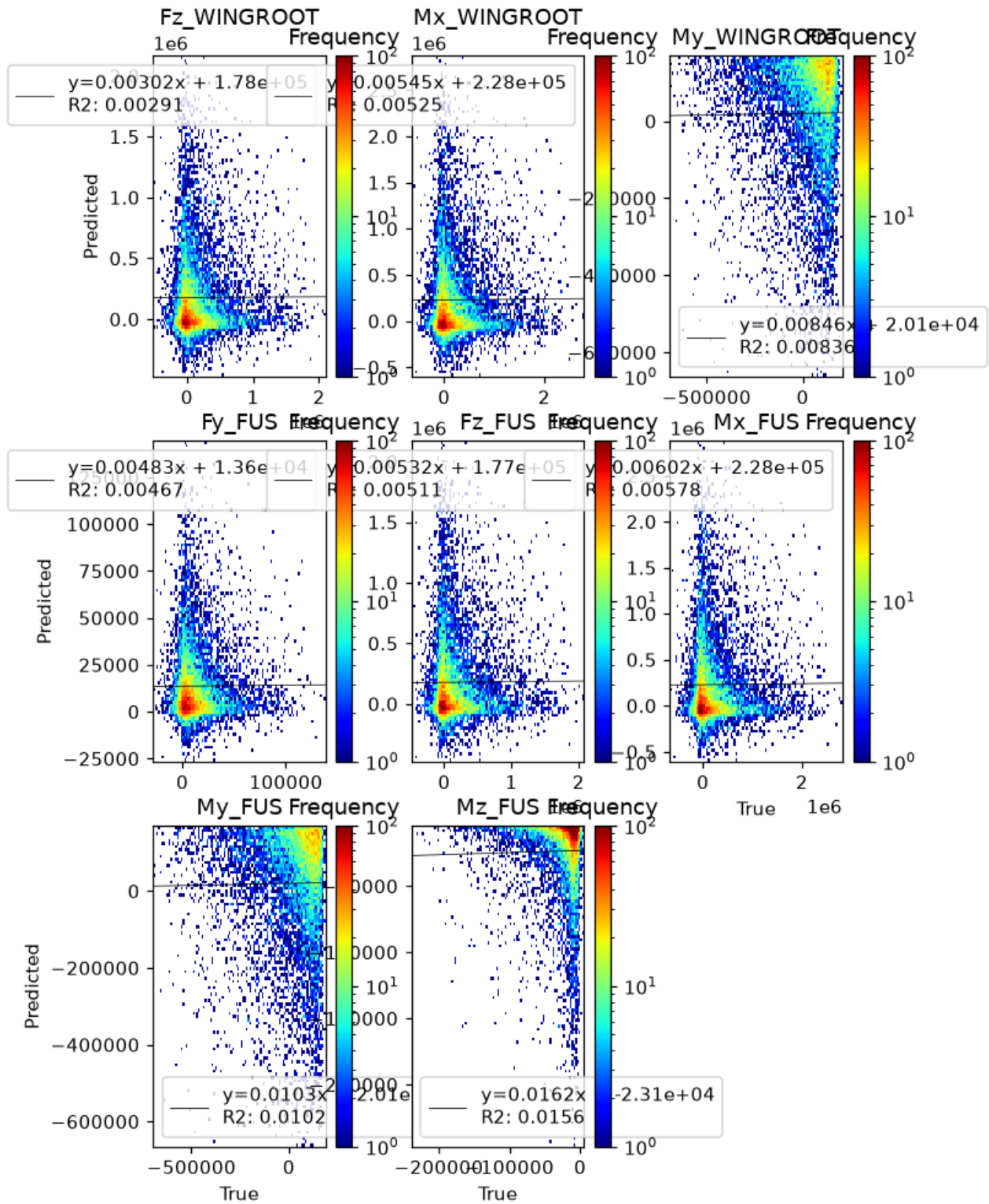


|             | AD     | KS       |
|-------------|--------|----------|
| Fz_WINGROOT | 0.0043 | 0.0022   |
| Mx_WINGROOT | 0.0012 | 0.000529 |
| My_WINGROOT | 0.001  | 7.13e-05 |
| Fy_FUS      | 0.0337 | 0.0397   |
| Fz_FUS      | 0.231  | 0.251    |
| Mx_FUS      | 0.25   | 0.4463   |
| My_FUS      | 0.0355 | 0.2169   |
| Mz_FUS      | 0.001  | 1.33e-55 |

True vs predicted distribution tests. Red:  $p < 0.05$ .

## True vs Predicted — 2D Histogram

Log-scaled frequency with a linear fit; the normalized slope should be close to 1.



## P(E|X) — Error Conditional on Inputs

We study whether the model error depends on the input variables. Two approaches are used: (1) a Kruskal-Wallis (KW) test on binned inputs (Sturges rule for bin count), and (2) violin plots showing the error distribution within each input bin.

Significant input-conditional bias ( $p < 0.05$ ) detected for: **Mach→Fz\_WINGROOT, Mach→Mx\_WINGROOT, Mach→My\_WINGROOT, Mach→Fy\_FUS, Mach→Fz\_FUS, Mach→Mx\_FUS, Mach→My\_FUS, Mach→Mz\_FUS, Altitude\_ft→Fz\_WINGROOT, Altitude\_ft→Mx\_WINGROOT....** These inputs explain residual variance — consider polynomial features or interaction terms.

### Bias Detection Table (KW p-values)

Green ( $p \geq 0.05$ ) = residue uniform across input bins; red ( $p < 0.05$ ) = significant bias.

|                    | Fz_WINGROOT | Mx_WINGROOT | My_WINGROOT | Fy_FUS    | Fz_FUS    | Mx_FUS    | My_FUS | Mz_FUS    |
|--------------------|-------------|-------------|-------------|-----------|-----------|-----------|--------|-----------|
| Mach_binned        | 4.09e-156   | 1.83e-155   | 0           | 3.85e-121 | 2.51e-156 | 1.99e-158 | 0      | 2.66e-175 |
| Altitude_ft_binned | 8.46e-113   | 2.89e-110   | 0           | 3.66e-120 | 7.38e-114 | 1.36e-112 | 0      | 2.62e-146 |
| ALPHA_deg_binned   | 0           | 0           | 0.0151      | 0         | 0         | 0         | 0.0233 | 0         |

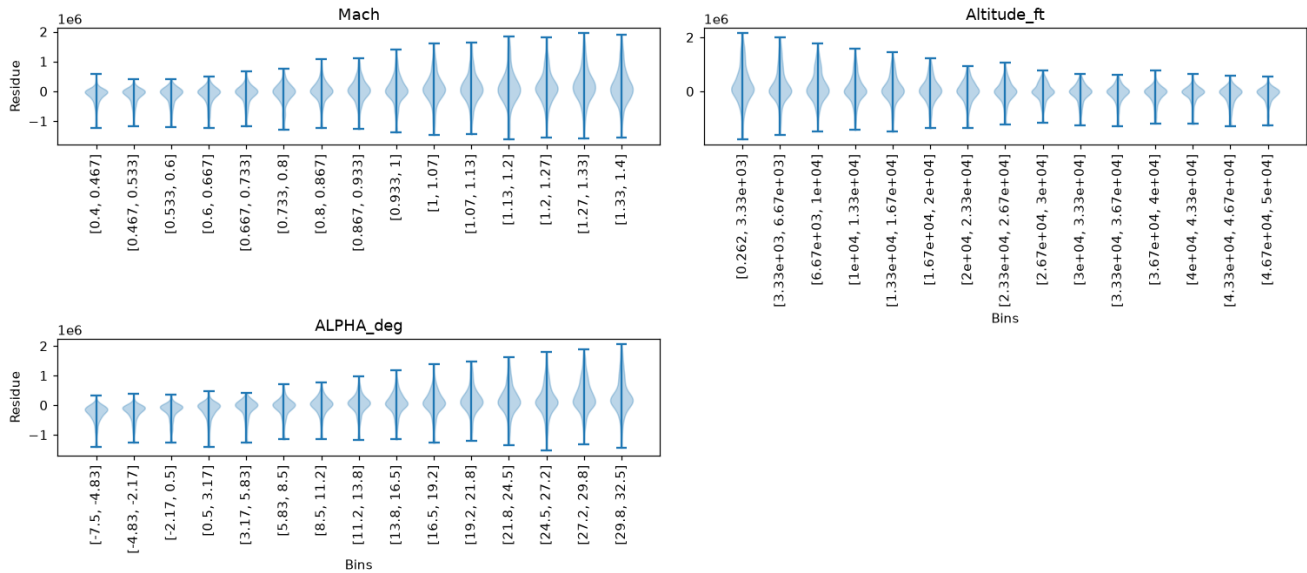
Kruskal-Wallis p-values on Sturges-binned inputs. Red:  $p < 0.05$  (error depends on that input).

## Residue Violin Plots per Output (binned by input)

Each panel shows the residue distribution for each input bin. A flat median line (dashed at 0) and symmetric violins indicate absence of bias.

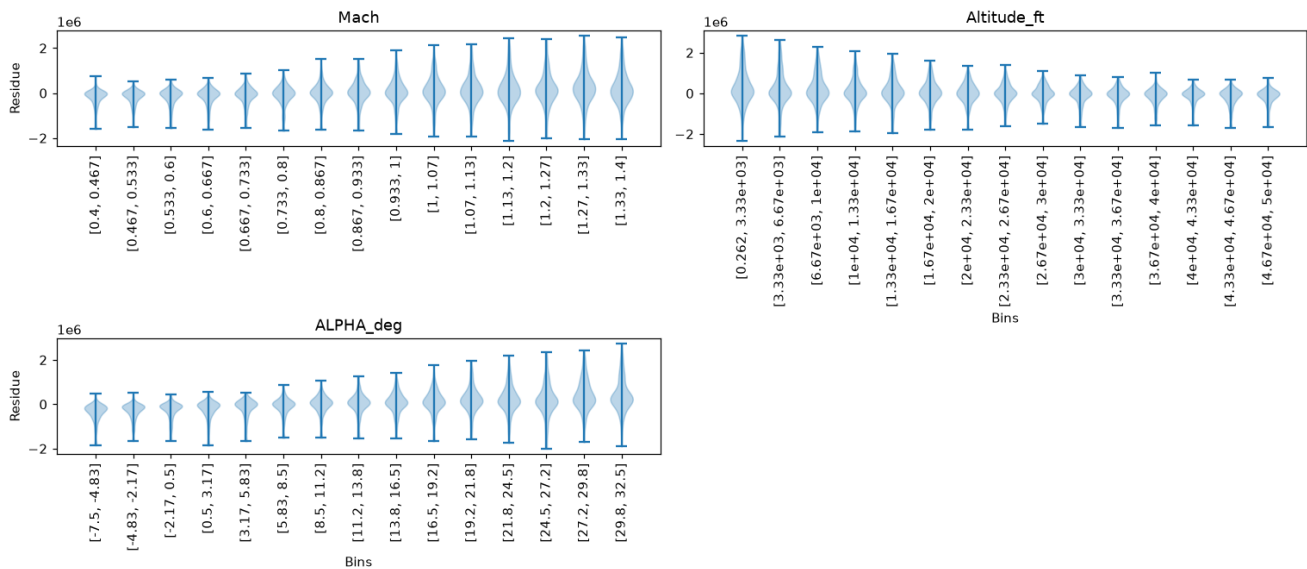
Residue — Fz\_WINGROOT

**Fz\_WINGROOT**  
Violin plot [Bins vs Residue] inside  $\mu \pm 3\sigma$



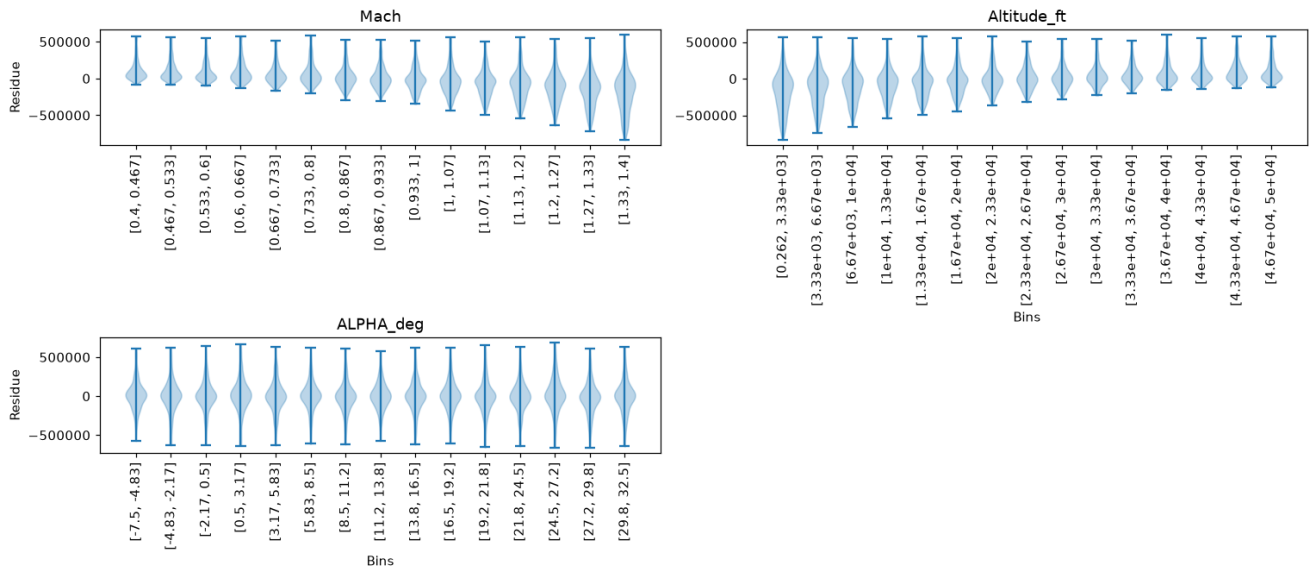
Residue — Mx\_WINGROOT

**Mx\_WINGROOT**  
Violin plot [Bins vs Residue] inside  $\mu \pm 3\sigma$



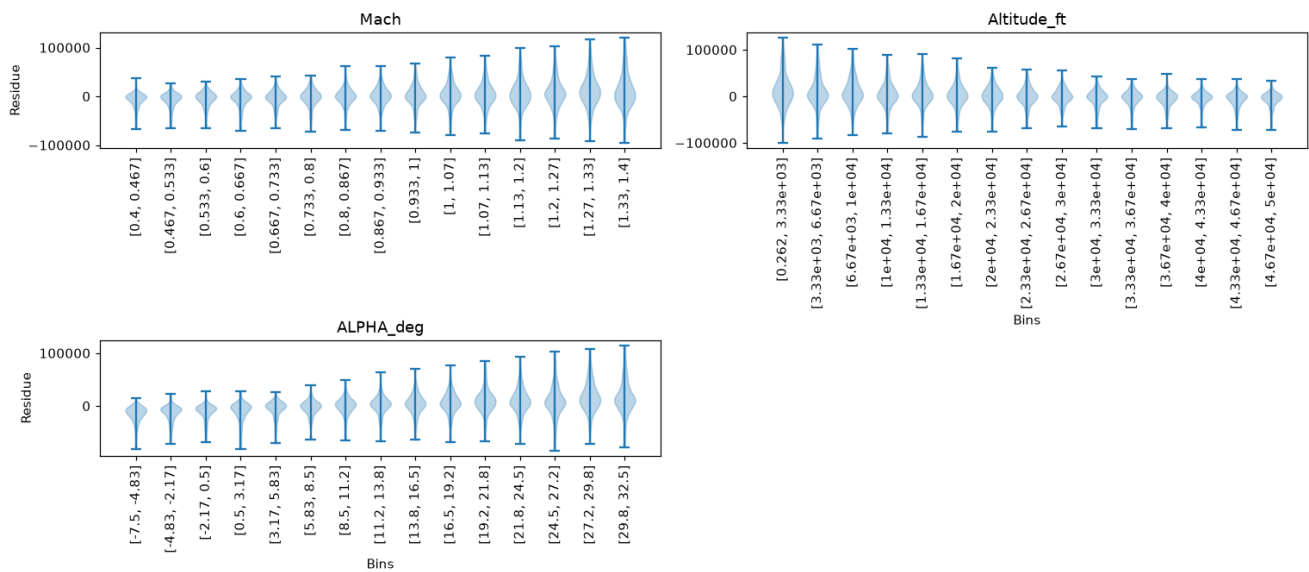
Residue — My\_WINGROOT

My\_WINGROOT  
Violin plot [Bins vs Residue] inside  $\mu \pm 3\sigma$



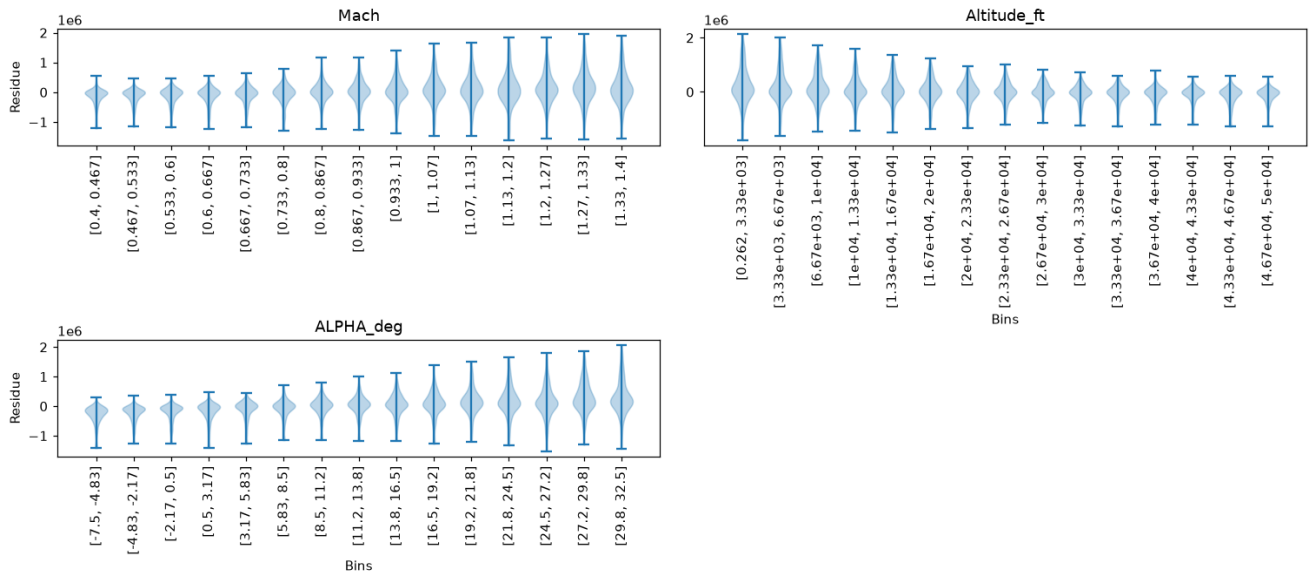
Residue — Fy\_FUS

Fy\_FUS  
Violin plot [Bins vs Residue] inside  $\mu \pm 3\sigma$



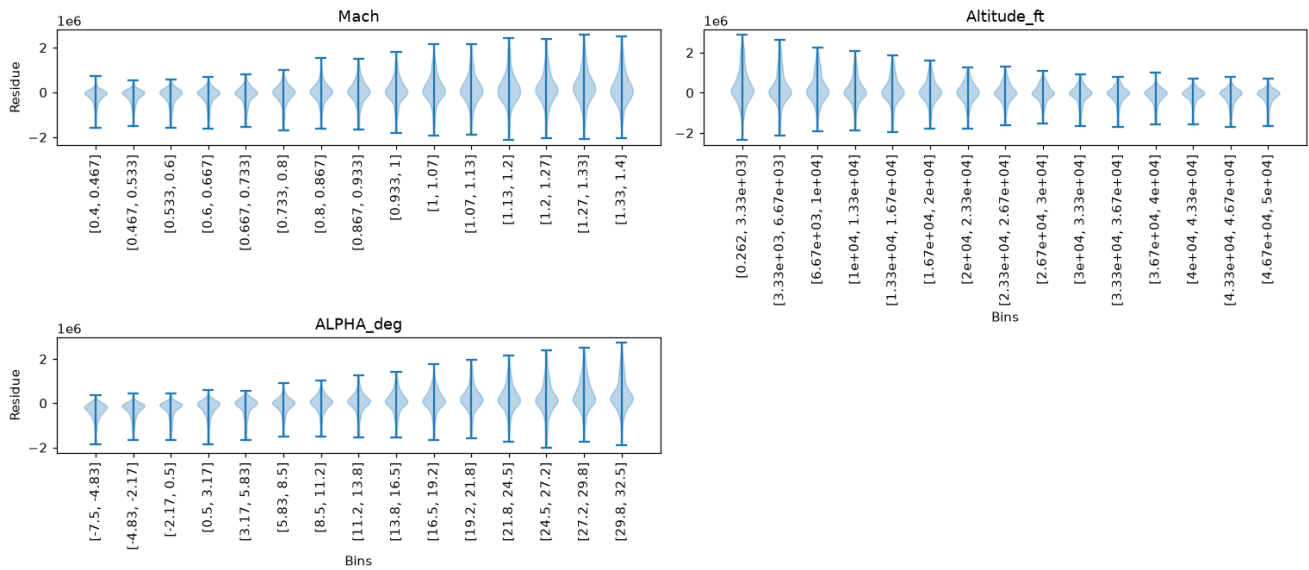
Residue — Fz\_FUS

**Fz\_FUS**  
Violin plot [Bins vs Residue] inside  $\mu \pm 3\sigma$



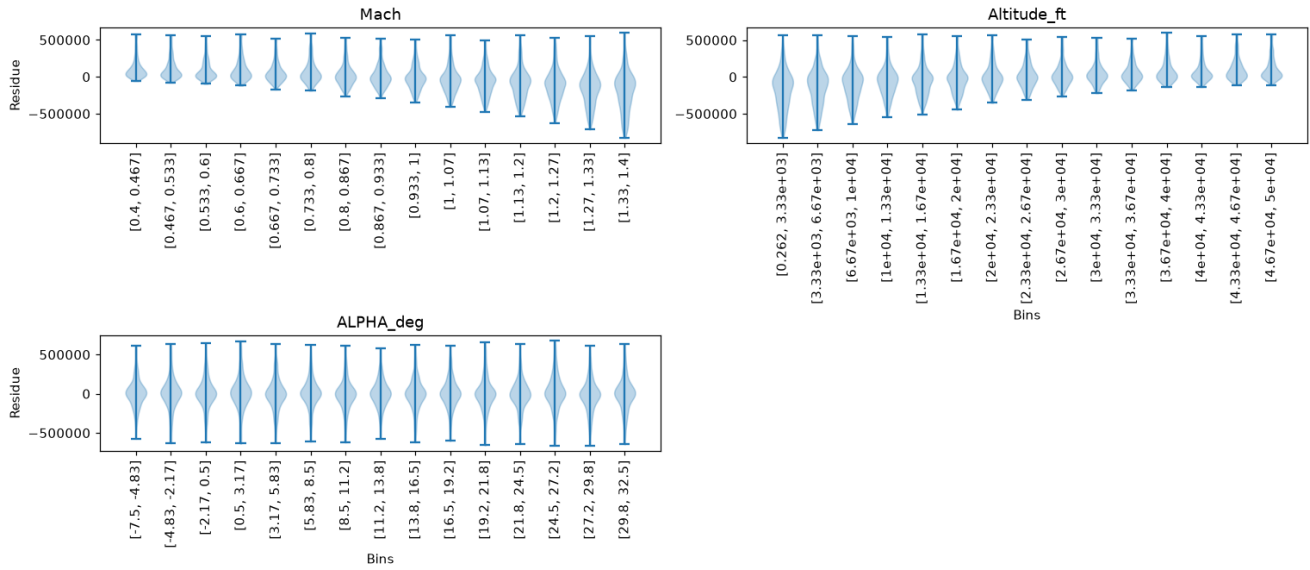
Residue — Mx\_FUS

**Mx\_FUS**  
Violin plot [Bins vs Residue] inside  $\mu \pm 3\sigma$



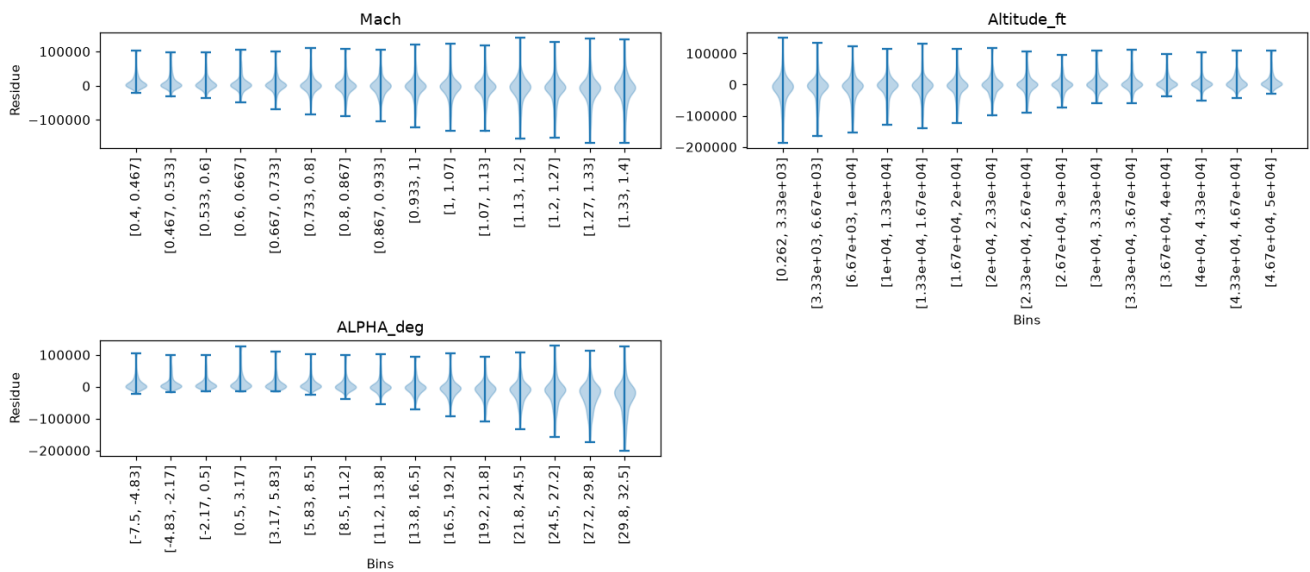
Residue — My\_FUS

My\_FUS  
Violin plot [Bins vs Residue] inside  $\mu \pm 3\sigma$



Residue — Mz\_FUS

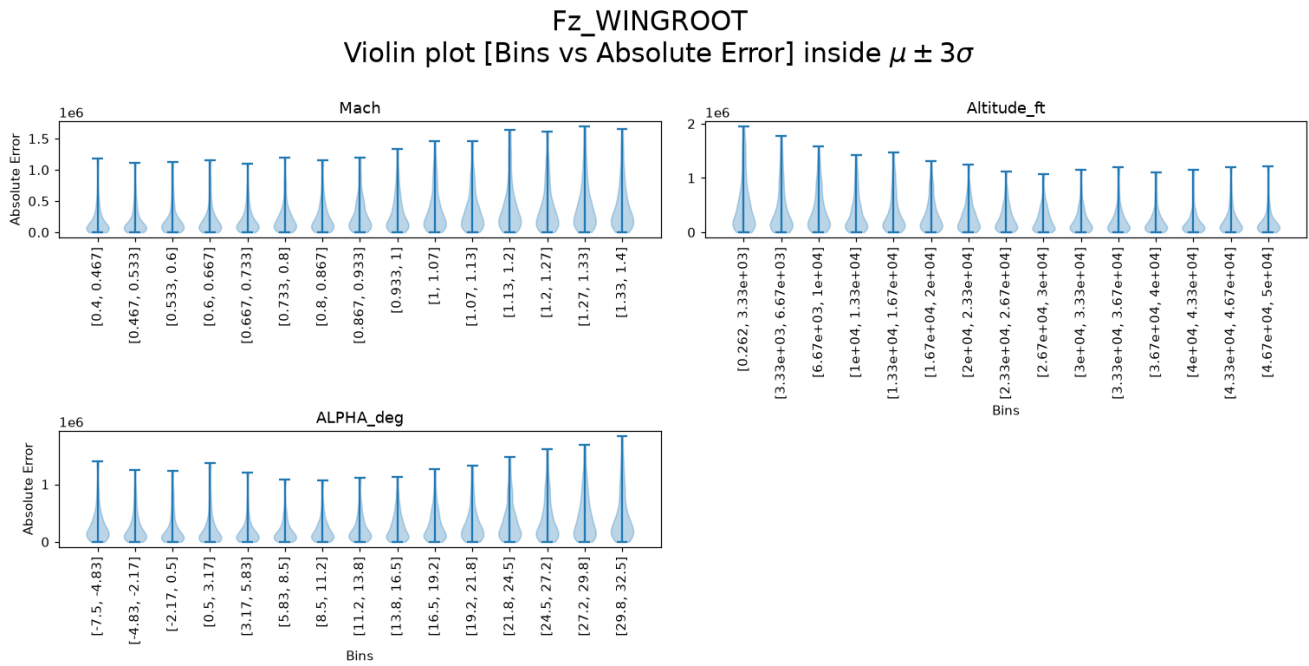
Mz\_FUS  
Violin plot [Bins vs Residue] inside  $\mu \pm 3\sigma$



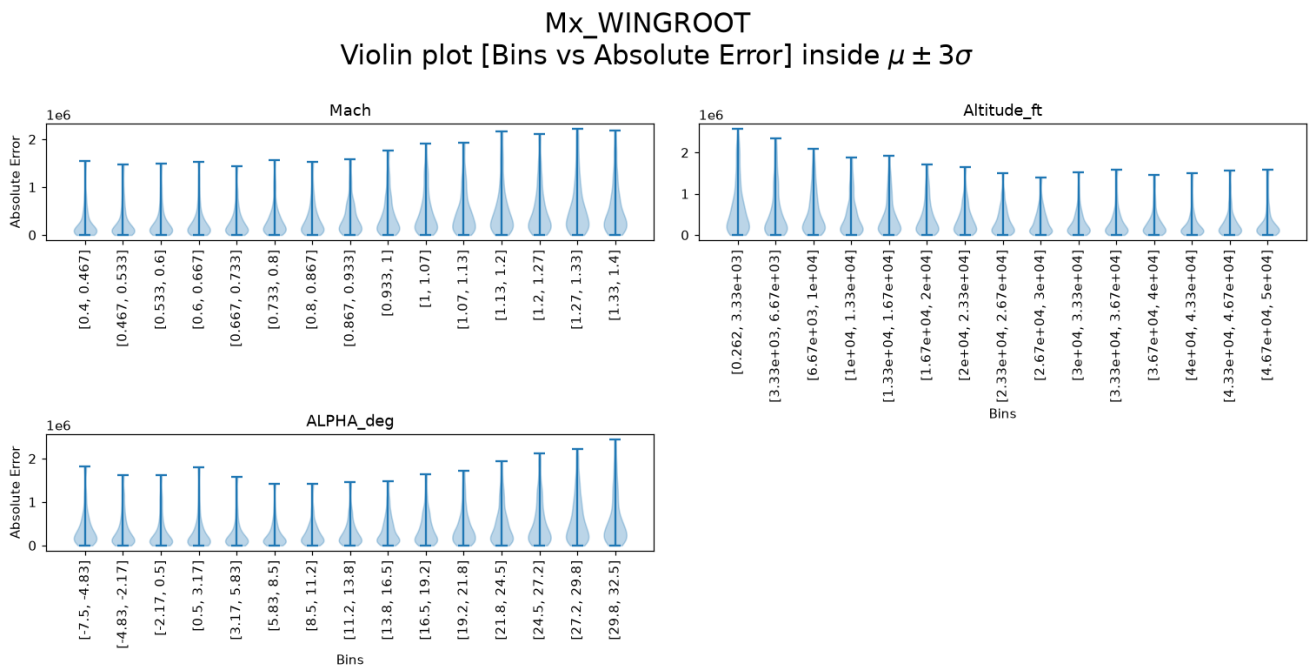


## Absolute Error Violin Plots per Output (binned by input)

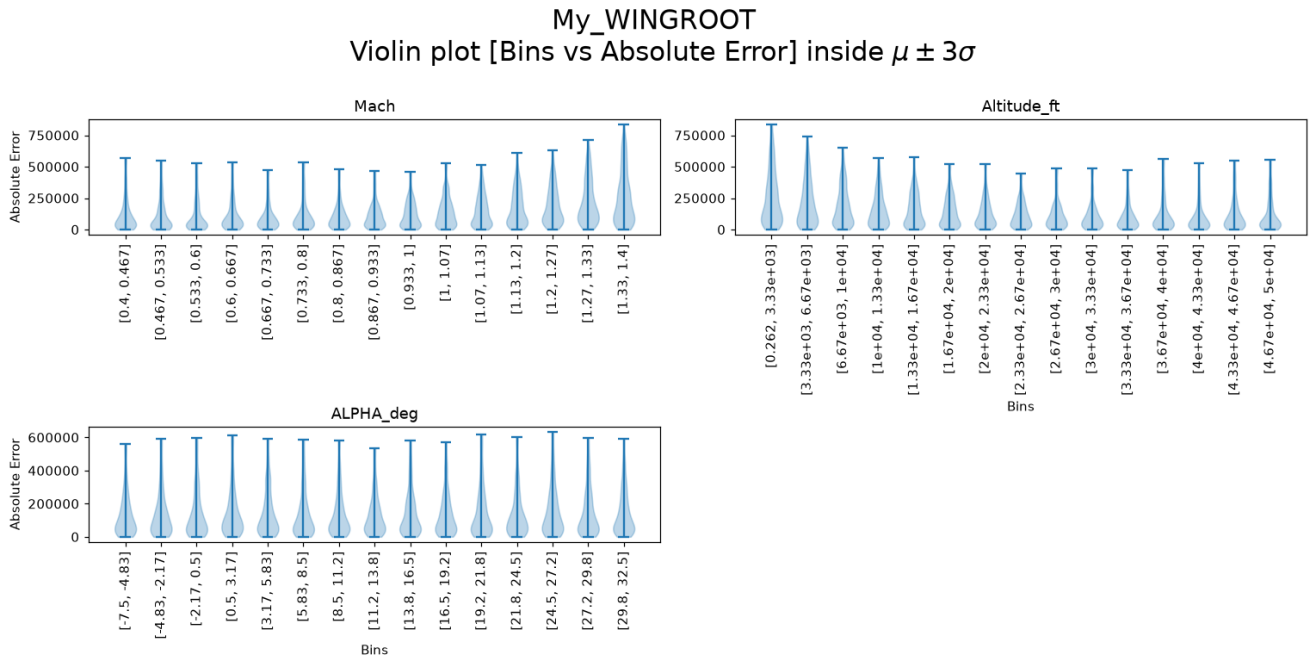
## Absolute Error — Fz\_WINGROOT



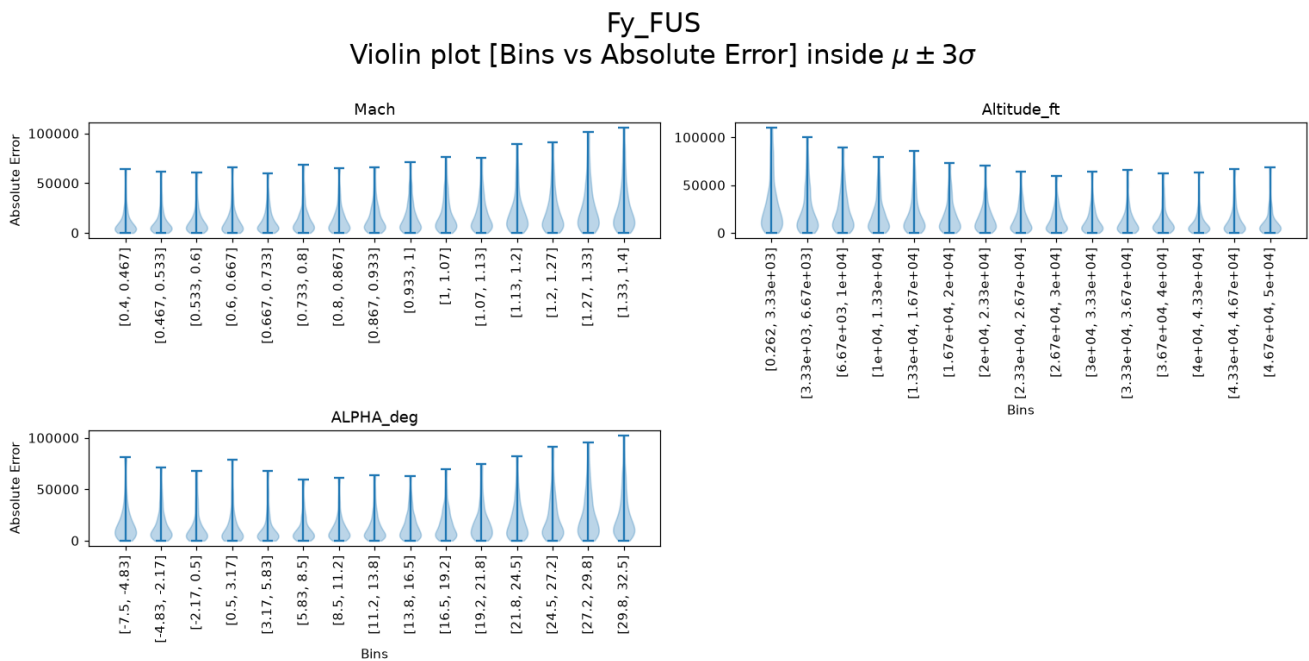
## Absolute Error — Mx\_WINGROOT



Absolute Error — My\_WINGROOT

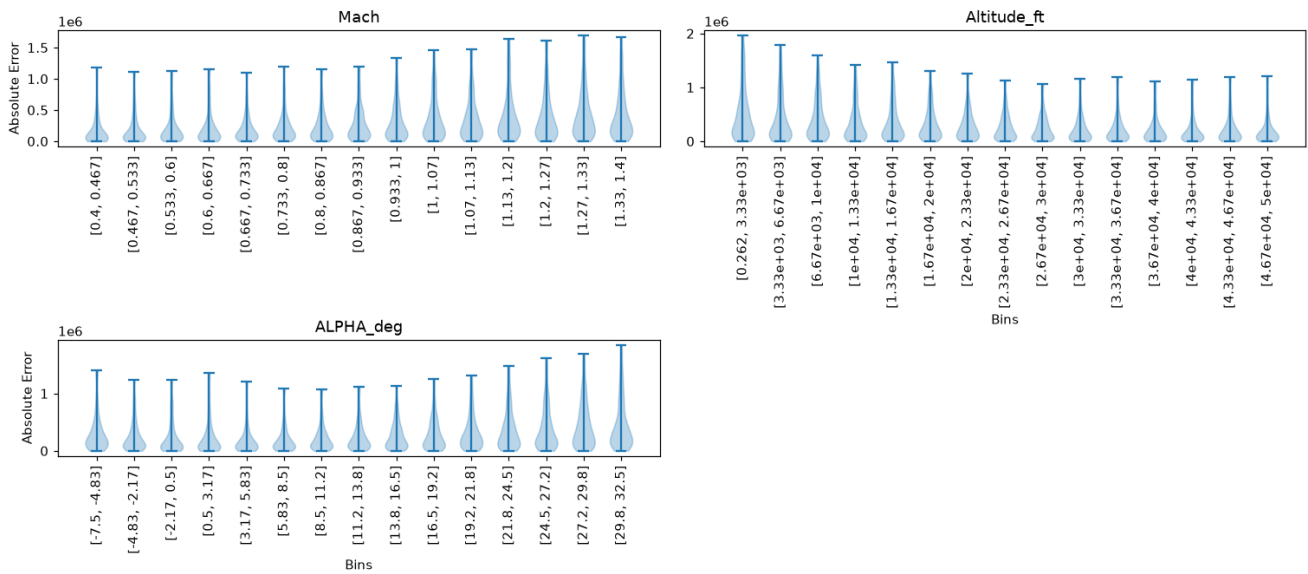


Absolute Error — Fy\_FUS



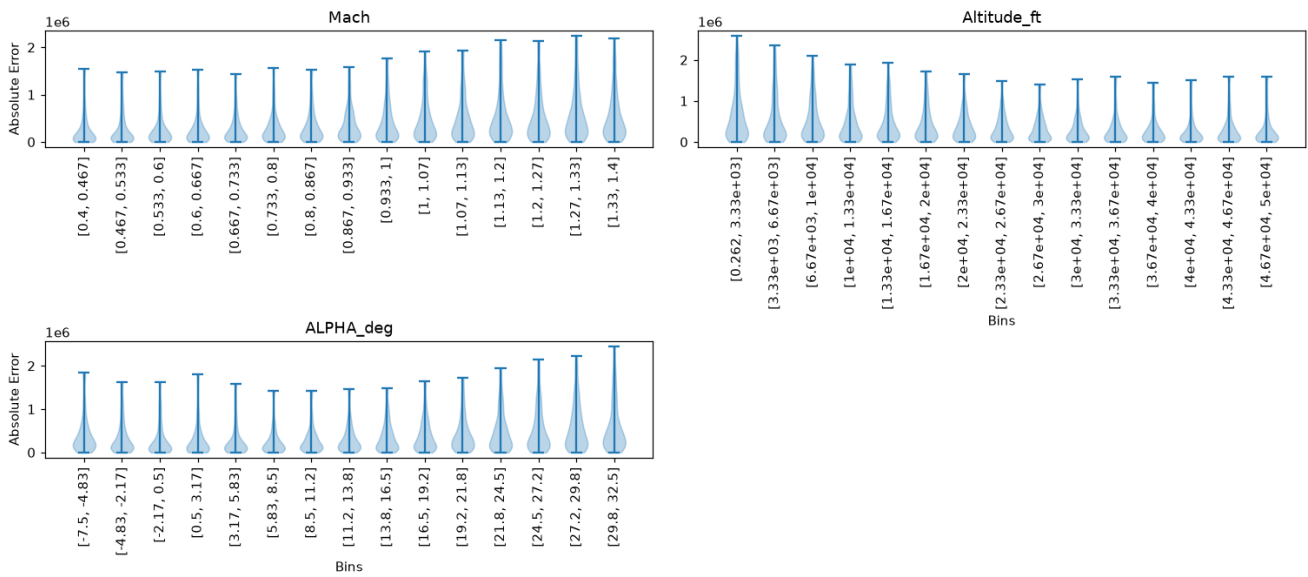
## Absolute Error — Fz\_FUS

**Fz\_FUS**  
Violin plot [Bins vs Absolute Error] inside  $\mu \pm 3\sigma$

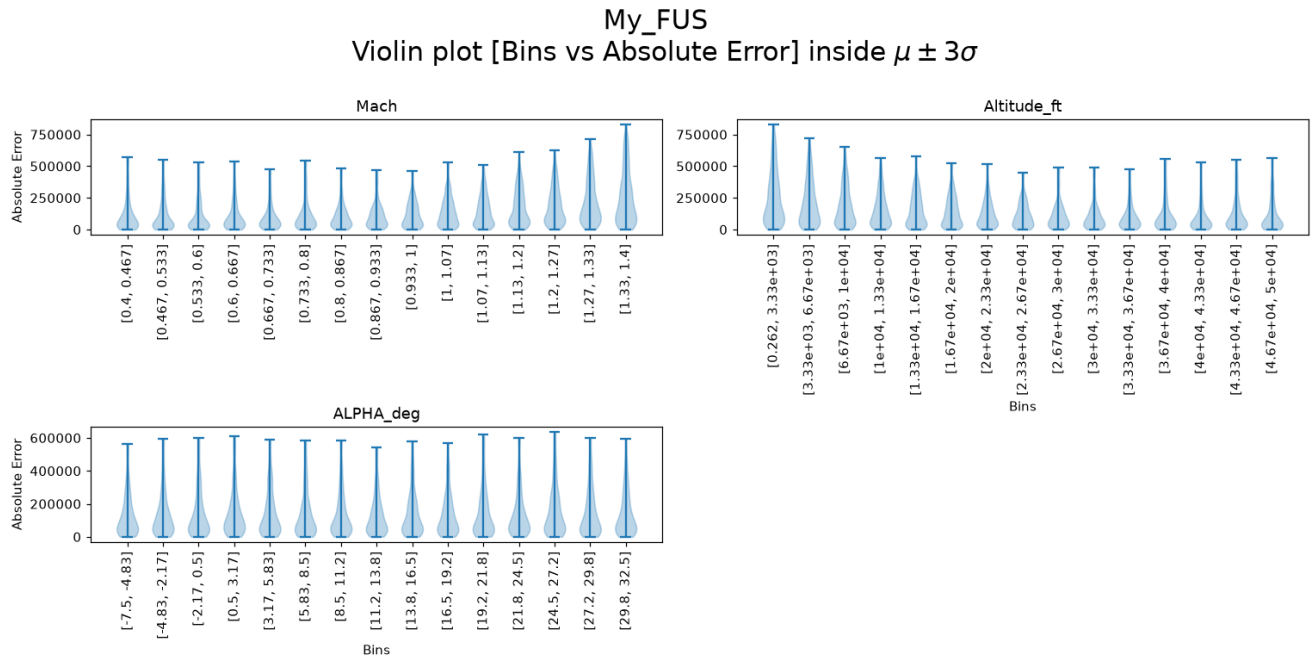


## Absolute Error — Mx\_FUS

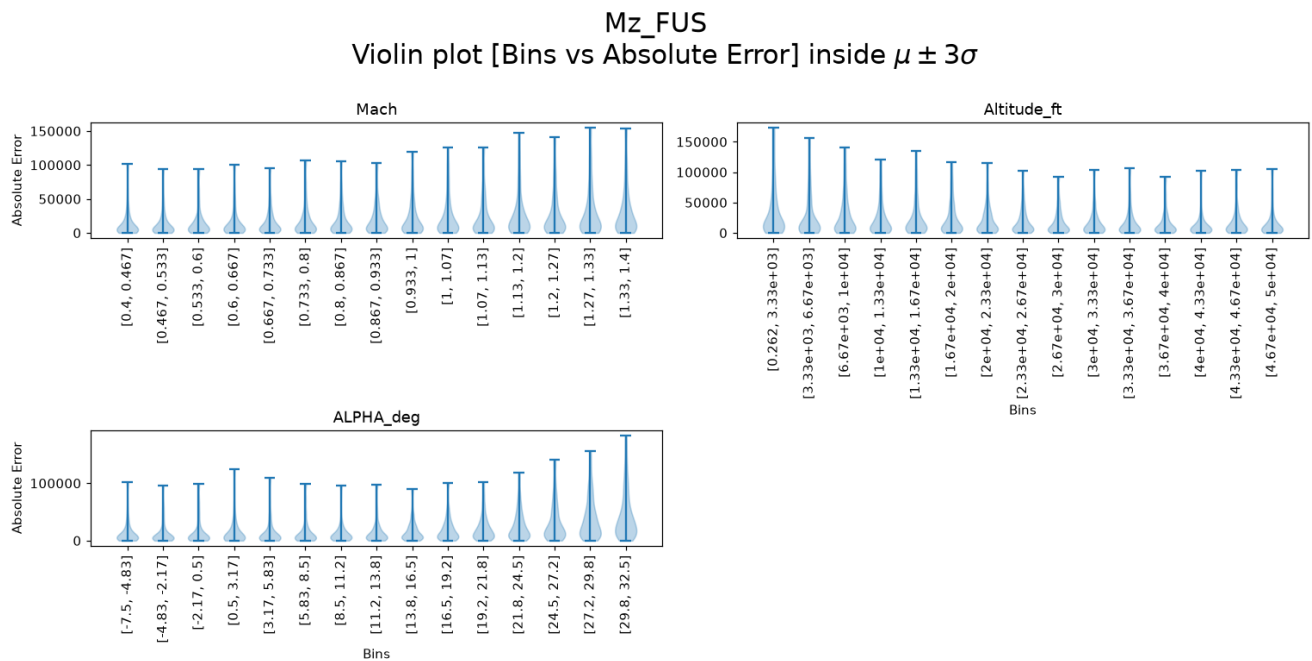
**Mx\_FUS**  
Violin plot [Bins vs Absolute Error] inside  $\mu \pm 3\sigma$



Absolute Error — My\_FUS



Absolute Error — Mz\_FUS



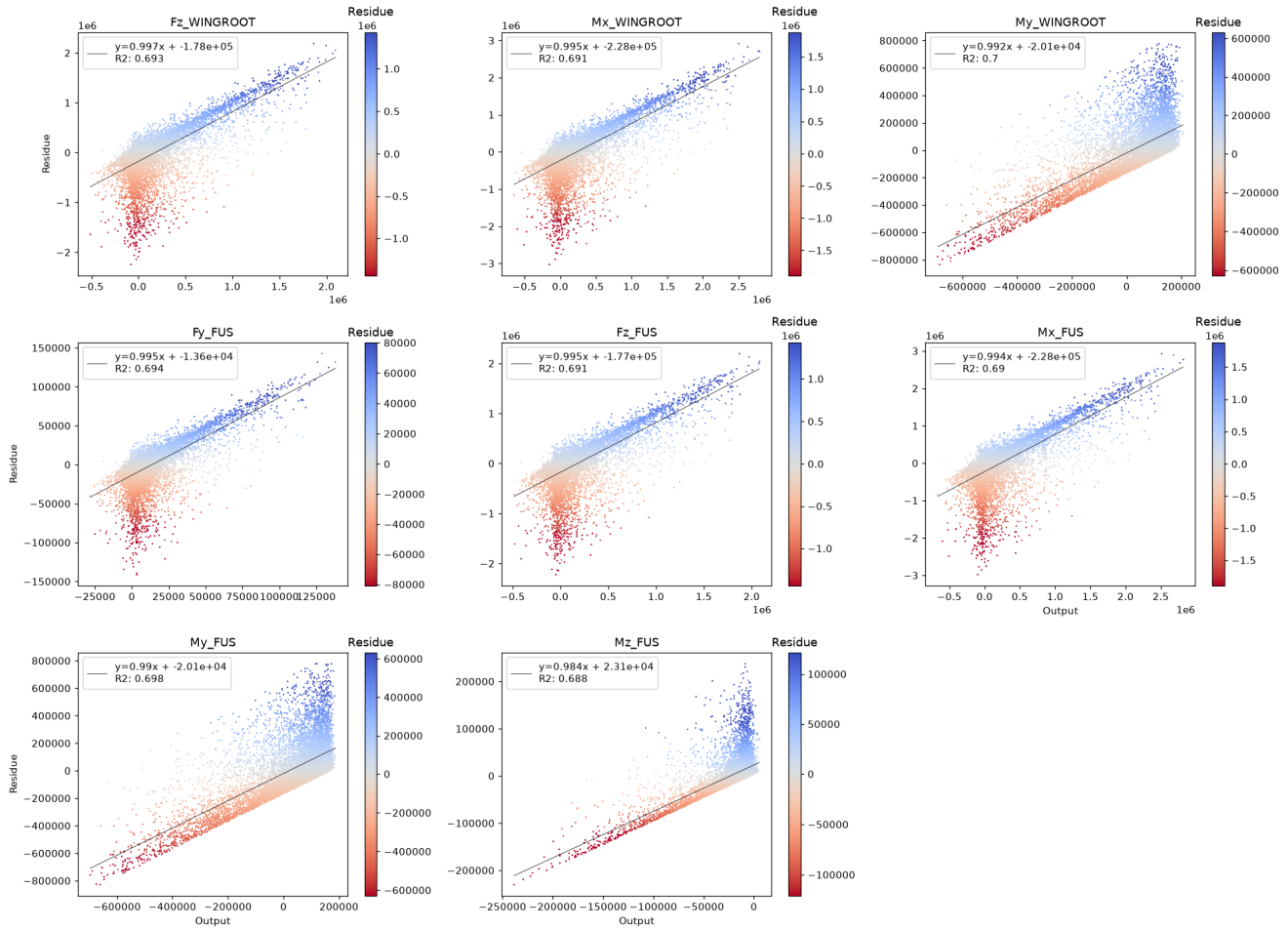
## P(E|Y) — Error Conditional on Outputs

We study whether the error depends on the output magnitude. A significant trend (Pearson or Spearman  $p < 0.05$ ) means the model is more accurate in some output ranges than others — a form of heteroscedasticity.

### Residue vs True Output — Scatter

Pearson  $r$  and  $p$ -value shown in title. Red title = significant linear trend.

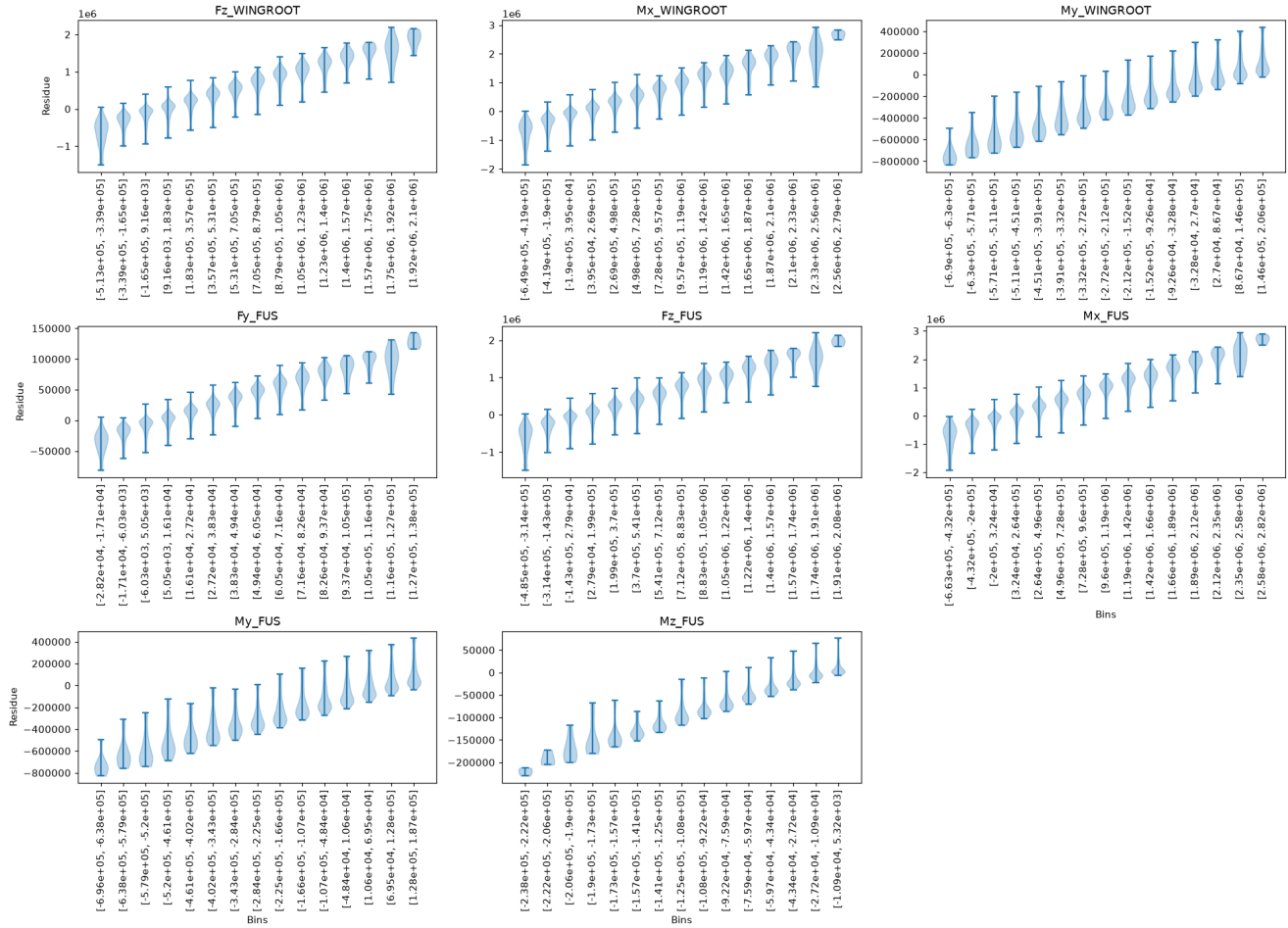
Scatter plot [Output vs Residue] (color trimmed to  $\mu \pm 3\sigma$ )



## Residue Violin vs True Output Bins

Bins determined by Sturges rule. Median line dashed at 0.

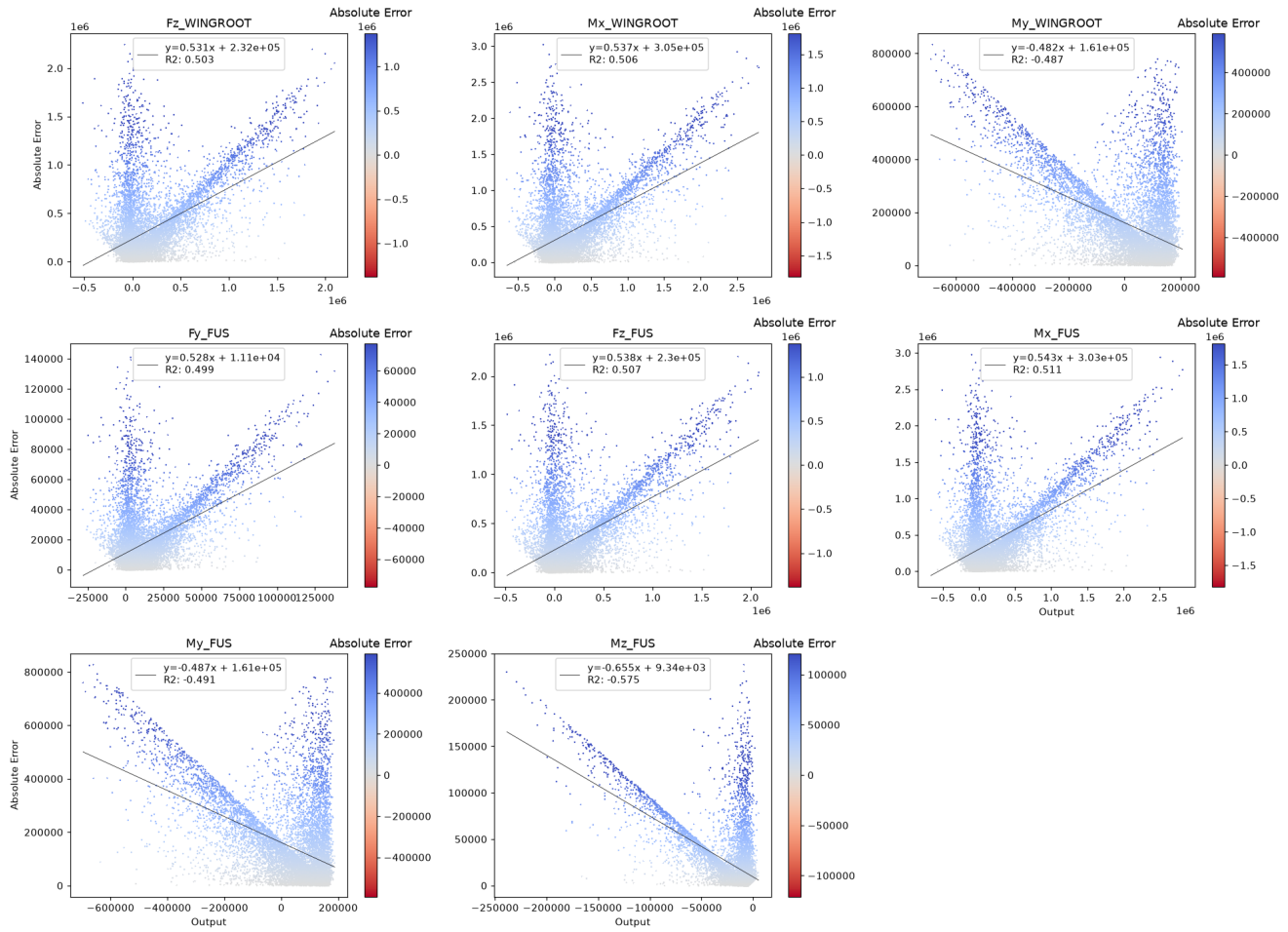
Violin plot [Bins vs Residue] inside  $\mu \pm 2\sigma$



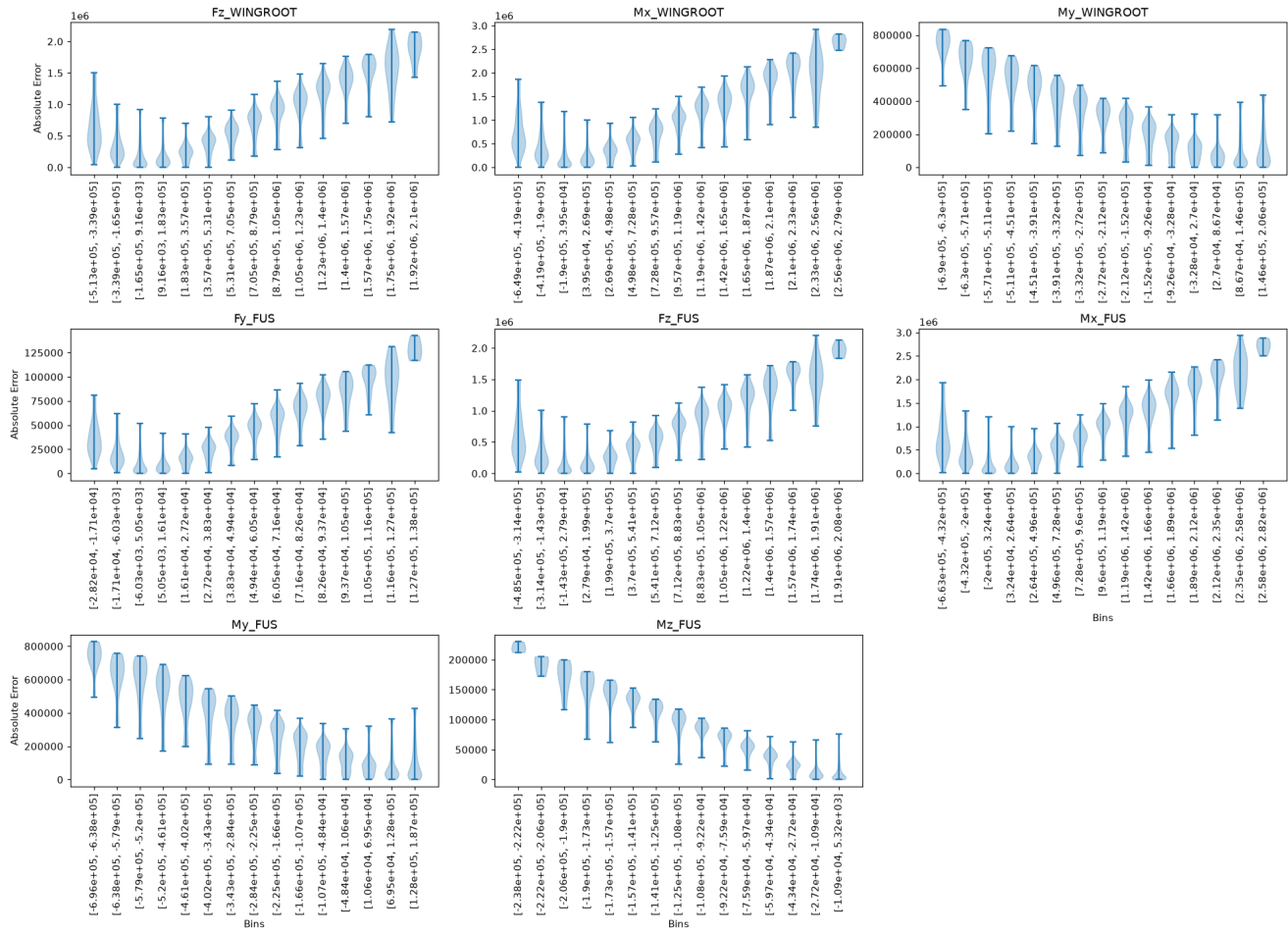
## Absolute Error vs True Output — Scatter

Spearman r shown. Red title = significant monotonic trend.

Scatter plot [Output vs Absolute Error] (color trimmed to  $\mu \pm 3\sigma$ )



## Absolute Error Violin vs True Output Bins

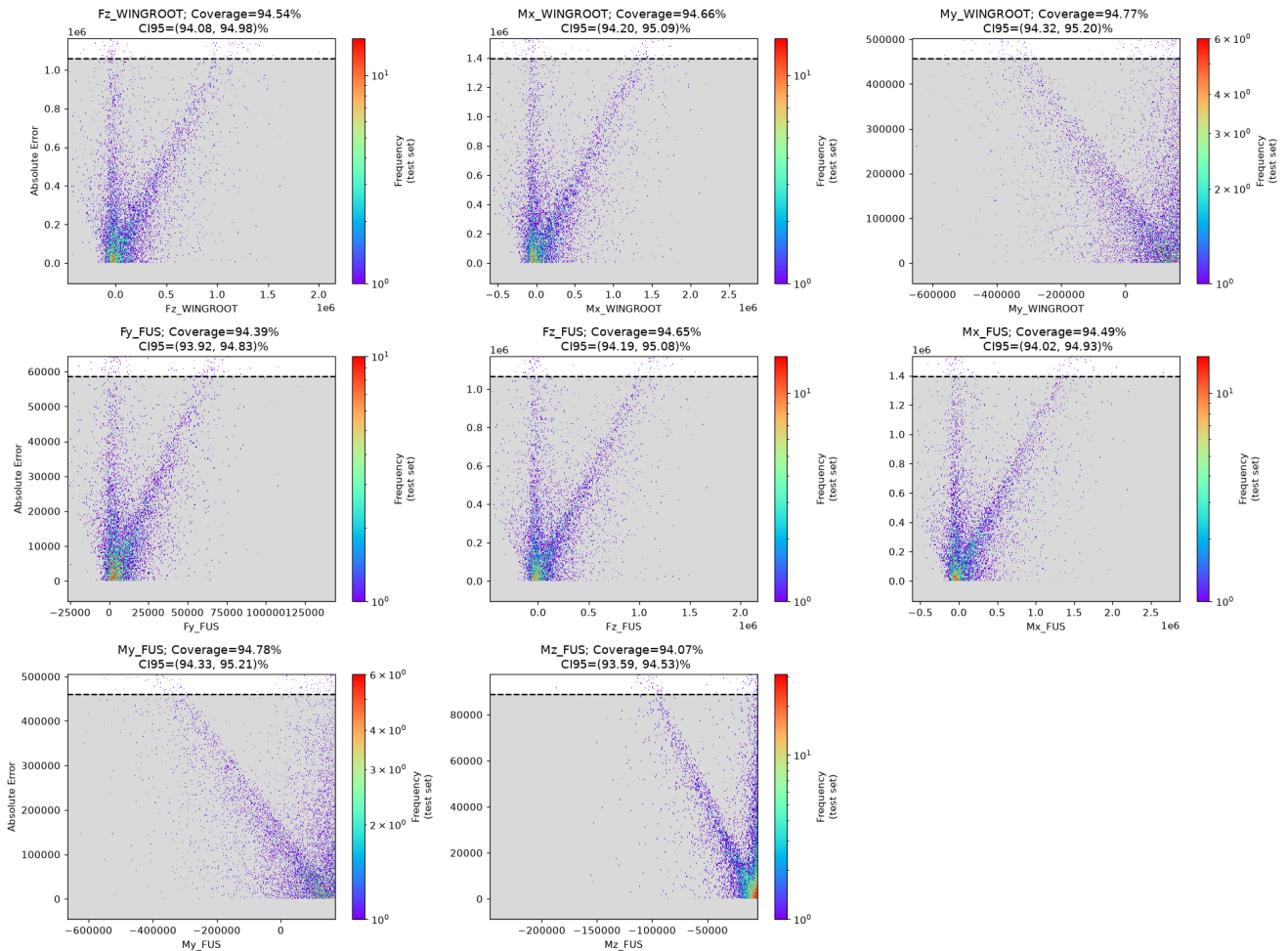
Violin plot [Bins vs Absolute Error] inside  $\mu \pm 2\sigma$ 



## Uncertainty Analysis

A global binned uncertainty model (95th percentile, right-tailed) is trained on the validation split of the absolute error and evaluated on the held-out test split, as in the extended validation notebook. Coverage is the proportion of test points whose error falls inside the predicted bound; it should be close to 95%.

### Coverage Plot



### Coverage Table

|          | Fz_WINGROOT    | Mx_WINGROOT   | My_WINGROOT   | Fy_FUS         | Fz_FUS         | Mx_FUS         | My_FUS         | Mz_FUS         | TOTAL COV |
|----------|----------------|---------------|---------------|----------------|----------------|----------------|----------------|----------------|-----------|
| Coverage | 94.54          | 94.66         | 94.77         | 94.39          | 94.65          | 94.49          | 94.78          | 94.07          | —         |
| CI       | [94.08, 94.98] | [94.2, 95.09] | [94.32, 95.2] | [93.92, 94.83] | [94.19, 95.08] | [94.02, 94.93] | [94.33, 95.21] | [93.59, 94.53] | —         |

Empirical coverage of the 95th-percentile uncertainty bound on the test split.